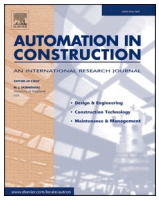
Automation in Construction 147 (2023) 104728



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Intelligent control of building fire protection system using digital twins and semantic web technologies

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ARTICLE INFO

Keywords:
Digital twins (DTs)
Sematic web
Ontology
Building automated system
Fire protection system
Intelligent control
Building information modelling (BIM)

ABSTRACT

A fire protection system takes on a critical significance to building operation. This paper describes the use of digital twins (DTs) and semantic web technologies for the intelligent control of building fire protection (BFP) systems in fire accidents. Specifically, a data fusion stage and several information-based control mechanisms are involved in the use of the two technologies. A designed BFP ontology is considered as the semantic model and basis of data fusion between the static building geometric information and the dynamic sensing data. The above information is incorporated into a DT data model, which is considered as the mapping of physical space. Moreover, rule models and process models are developed to achieve intelligent control mechanisms and keep the DT data model synced with the physical space. A case study based on a fire accident simulation was conducted to verify the feasibility of the use of DTs and semantic web technologies.

1. Introduction

Fire protection has become a vital research topic in the field of public safety protection since there have been considerable property damage arising from fire incidents and fire casualties. Building fire protection (BFP) system is capable of responding quickly to a fire emergency, which is of great significance to reduce losses. In conventional fire alarm systems, human reactions are required to inform the fire department to track down the fire location and organise an evacuation after a fire incident is detected [1]. In most cases, if people are not alerted in time or fail to control the fire at its early stages, the fire will be more serious and cause more losses [2]. For instance, the delayed evacuation of the Grenfell Tower Fire in London in June 2017 was cited as a great tragedy, which gutted the entire building, killing 80 people and injuring over 70 people [3,4]. In addition, insufficient information regarding the fire incident may hinder the organisation of evacuation and rescue [5]. Accordingly, there is an urgent need for an intelligent fire protection system that can display information regarding fires and provide rational strategies to deal with fire incidents automatically at an early stage.

An intelligent fire protection system refers to a comprehensive building automation system that comprises several basic subsystems (e.g., fire alarm systems, smoke control systems, fire door monitoring systems, sprinkler systems, emergency broadcast systems, as well as evacuation guidance systems). Intelligent fire protection entails activating the fire alarm system, analysing the current fire situation, and then sending commands to other subsystems to respond to fire emergencies. In the above processes, timely feedback and integration of information are of high importance. The application of some emerging technologies in this regard has been investigated extensively. For example, building information modelling (BIM) provides a three-dimensional (3D) view of a building and stores and retrieves the information regarding the building. The vivid 3D environment created using BIM may provide visual guidance for fire rescue operations [6]. Furthermore, the Internet of Things (IoT) is an integration of technologies that creates connections between all information sensing devices and the internet to support effective sensing, reliable transmission, and centralised management of data collected by IoT end products [7]. The integration of BIM and IoT systems is capable of presenting the real-time state and the as-built structures of a building. Fang et al. [5] developed a framework of an IoT-aided system that can integrate real-time monitoring data of the fire environment and people counting collected by multiple BIM detectors and sensors to dynamically optimize building evacuation. Moreover, the integration of BIM, IoT, virtual reality (VR), and augmented reality (AR) can underpin path finding in fire rescue

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https://doi.org/10.1016/j.autcon.2022.104728
Received 26 May 2022; Received in revised form 15 December 2022; Accepted 19 December 2022
Available online 30 December 2022
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[8,9]. In brief, the above existing technologies have the potential to increase BFP efficiency. However, since recorded data are collected in different formats from a wide variety of sources, and the design of building structures and IoT networks are complicated, data interoperability and system coordination pose critical challenges to intelligent control of the fire protection system during fire emergencies.

Digital twins (DTs) are considered as an applicable solution to the above challenges. The emergence of the DT has increased efficiency in numerous industries. DT's applied research in the architecture, engineering, and construction (AEC) domain has aroused wide interest since 2018 [10]. The DT-driven approach is capable of facilitating the management, planning, forecasting, and presentation of buildings, infrastructures, or city assets [11]. Since DT enables the integration of virtual representation and real-time data, dynamic monitoring and predictive operation can be achieved through facilities management [12,13]. The predictive approach can reduce operational cost and time loss. On that basis, the fire scene information collected from the IoT systems can be integrated with BIM models to provide a digital representation of the fire protection system. Data analysis and emergency simulation can be achieved using the virtual fire protection system. Thus, DT serves as the real world's intelligent brain to coordinate various subsystems to cope with fire incidents. The application of DT to the intelligent control of BFP systems comprises two crucial aspects as follows: (1) the digital representation is a high-precision mapping of real-world objects; (2) dynamic updating mechanisms are essential to ensure that the virtual is always in sync with the real. However, previous research has placed a major focus on anomaly detection or prediction based on deep learning methods. Research on the interactive feedback between the virtual and the real is still required, which takes on a great significance to the intelligent control of the fire protection system. Furthermore, as the requirements for building information vary throughout the lifecycle and the data are in different formats (e.g., building models, videos, pictures, documents), a unified data description is expected to solve the issue of data exchange and reuse.

This study describes the use of DTs and semantic web technologies for the intelligent control of BFP systems. Data are organised as a linked graph based on the Resource Description Framework (RDF) [14], such that a uniform representation syntax is provided for building data in different formats. Accordingly, the query language, SPARQL Protocol and RDF Query Language (SPARQL) [15], can be adopted to retrieve building information efficiently [16]. Moreover, ontology is a common vocabulary and formal description [17] to define how the above data are linked, and how computers understand the above data, such that the semantic interoperability between the data can be addressed. In the AEC domain, since a unified data integration environment is created, the use of semantic web technologies has been studied as a promising strategy for data interoperability [18,19], information exchange [20,21], and complex cross-domain linkage [22]. A series of works on the development of ontologies (e.g., IoT sensor observations ontology (SSN) [23], brick ontology [24], and building topology ontology (BOT) [25]) have laid a basis for the integration of BIM data and IoT data. In this study, an ontology is discussed for data fusion based on the above ontologies, and a rule model and a process model are built to achieve the interaction mechanism between the virtual and the real.

The rest of this study is organised as follows. In Section 2, recent literature on the application of BIM and IoT in fire risk management, DT modelling, and semantic web technologies used for data interoperability in the AEC domain is reviewed to identify the research gap. In Section 3, the use of DT and sematic web technologies is described for the intelligent control of the fire protection system. Moreover, a brief introduction of the critical procedure of the use of the two technologies for the intelligent control of BFP systems in fire accidents is given in this section. Section 4 verifies the feasibility and efficiency of the use of the two technologies with a case study, and the discussion is given in Section 5. Section 6 draws a conclusion of the study and elaborates on future work.

2. Literature review

2.1. Fire risk management: application of BIM and IoT

Fire risk management is of great significance to building operations that comprise fire protection design, fire/smoke spread limitation, fire extinguishing, and escape probability [26]. It is a complicated task, and there are several factors for the efficiency of fire risk management, which include the characteristics of the occupants, the building facilities, and the features of the fire [27]. An integrated environment capable of managing and analysing the above factors should be established [28]. The use of BIM has shown its potential as such a solution.

From the perspective of modelling, BIM can be considered as a 3D model with considerable building information. The geometric data of BIM are beneficial to the assessment of the reliability of the design of fire protection facilities [28], planning escape routes [29,30], and demonstration of the real-time fire status [31]. In addition, the informatization features of BIM (e.g., information storage, extraction, and retrieval) can be employed as the database for BIM-based fire safety information management. For instance, Wang et al. [32] applied BIM to support asset management for fire safety (e.g., the maintenance information of fire safety equipment). Wu and Ma [33] proposed a BIM-based fire emergency management system that comprises a fire intelligence monitoring module, a fire warning module, a fire response module, as well as a fire treatment module. The proposed system is capable of acquiring fire information, determining fire status, and making decisions for trapped people. However, existing research has mainly focused on the fire protection design and escape probability, whereas a series of closed-loop control logic of fire sensing, fire/smoke spread control, fire extinguishing and evacuation security based on BIM has not been established. To achieve the above goals, BIM should be integrated with the fire protection system to achieve fire protection equipment management while enabling the fire protection system to respond promptly and rapidly in fire emergencies.

In general, the rapid response of fire protection systems comprises early detection and reasonable reactions. The former provides more time for escaping and extinguishing the fire by using the detectability of the terminals (e.g., the very early smoke detection apparatus [34]). In this study, the reasonable reaction of the fire protection system is primarily analysed, with a stress on fast actions after a fire incident is detected (e.g., activating fire extinguishing terminals, opening smoke systems, and guiding escape). Compared with conventional manual operation, the emergence of IoT, a network of smart devices, enables automated control of sensing terminals by providing an environment where the devices can communicate with and sense each other. Nevertheless, existing research has placed a focus on monitoring [35,36] and sensing data management [2]. Since BIM provides a 3D geometry model and building design information, and IoT enables data connection, communication, and collection from devices, the integration of BIM and IoT has been applied to road compaction quality monitoring and management [37], automatic fault detection and building diagnostics [13], as well as thermal comfort building assessment [38]. However, there has been rare research on fire risk control, especially on data-driven decision-making about the control of fire protection systems.

As a result, a methodology integrating BIM and IoT is proposed to develop an intelligent control strategy for BFP systems. The challenges originate from the realisation of the control mechanism and the interoperability between BIM data and IoT data [39]. Accordingly, a DT framework is built, and semantic web technologies are adopted to solve the above challenges.

2.2. Digital twin in AEC domain: from the modelling perspective

The concept of the DT originated from the concept ideal for product lifecycle management proposed by Grieves in 2003. Subsequently, the definition of DT has been enriched in the application of the aerospace

industry [40], smart manufacturing [41], infrastructure [42–44], and facility management [13]. Although the understanding of DT varies with application scenarios, the connotation has come to a consensus that DT stresses a digital representation of real-world objects and the connection between virtual representation and physical objects. Furthermore, model construction and model interaction are involved from the modelling perspective.

In the AEC domain, the DT model has been commonly constructed based on BIM. A light weighted BIM is employed to establish the DT-BIM hoisting planning model for prefabricated component hoisting management [45]. The model covers hoisting environment information (e.g., construction site layout), hoisting route information (e.g., hoisting point and installation point), and prefabricated component information (e.g., 3D shapes). At the city and building levels, BIM is considered as one of the digital models of physical city assets [46,47]. Since the Industry Foundation Classes (IFC) format gives a standardised and digital description of BIM, Stojanovic et al. [48] converted the 3D cloud data into IFC data to support the DT representation. Lu and Brilakis [49] developed a bridge geometric DT, comprising a wide variety of *IfcObjects*. Moreover, Lu et al. [13] adopted IFC as a centre data model of BIM and IoT data to achieve DT-based anomaly detection during daily building operation and maintenance (O&M). However, the above models are only concerned with the geometric information or operation information of the building objects.

The mechanism of the operation process takes on a great significance in describing the model interaction, and the mechanism is formed as various models according to the application scenarios. During building O&M processes, the mechanism of intelligent control is formed as a behaviour model and a rule model [50]. The behaviour model can make response and feasibility analyses for the manual operation and system directives; it comprises diagnostic rules, prediction rules, and decision rules. In safety risk control during prefabricated building hoisting operations, the safety risk mechanism was modelled as a Bayesian network to support decision-making [51].

A BFP system comprises several interoperating subsystems. Each of them implements a certain set of functions but also participates in the processes of handling fire emergencies. Process models are preferred to represent the procedures or workflows [52]. Accordingly, to express the system interoperating process and enable people to participate in the process, a process model is built based on the business process model and notation (BPMN) [53] to express the intelligent control mechanism of the fire protection system in this study.

2.3. Semantic web technologies for data interoperability in AEC domain

Data integration is recognized as a basic challenge of DT-based applications in practice. In the AEC domain, a plethora of data are obtained for a project due to the multiple phases involved in its lifecycle (e.g., design, construction, production, procurement, operation, and maintenance [10]). The above data may have a wide variety of modes (e.g., recording data, videos, figures, and building models). On that basis, the interoperability of the above heterogeneous data becomes the dominant issue in data integration. With the emergence of BIM, some studies have suggested that IFC, an open standard data schema of BIM, lays a basis for the data integration of building information, sensor data, and 3D point cloud data [13,48,49]. However, the complex nature of the IFC schema limits its extensibility, modularity, and cross-domain linking ability beyond geometric representation [54].

Semantic web technologies provide a paradigm shift for data integration across domains. It provides a common structure to represent data and a set of tools to support the manipulation, sharing, reuse, retrieval, and integration of heterogeneous data [55]. Data are modelled as a directed, labelled graph based on the RDF. The respective pair of nodes and their relationship, namely, the RDF triple, establishes a link between information from disparate data sources. Based on the RDF graph, SPARQL can be employed for sematic and relational searching.

Table 1
Brief summary of domain ontologies in AEC domain.

Ontology	Description	Reference
ifcOWL	A domain ontology for BIM information by mapping the EXPRESS schema to OWL [66].	https://technical.buildingsmart.org/standards/ifc/ifc-formats/ifcowl/
Building Topology Ontology (BOT)	BOT focused on core geometric concepts of an architecture and their topological relations.	https://w3c-lbd-cg.github.io/bot/bot.ttl
Building Product Ontology (BPO)	An ontology for describing building products,	http://www.w3id.org/bpo#
Semantic Sensor Network (SSN) Ontology	An ontology for describing sensors and related properties, including their observations, involved procedures, features, etc.	https://www.w3.org/ns/ssn/
Sensor, Observation, Sample, and Actuator (SOSA) Ontology	A lightweight ontology, which defines the elementary classes and properties for SSN ontology.	https://www.w3.org/ns/sosa/
DogOnt	An ontology to describe various types of building objects, such as devices, systems, building environments, architecture, etc.	http://elite.polito.it/ontologies/dogont.owl
Smart Appliances REFerence (SAREF) Ontology	An ontology that provides a shared description of existing assets in the smart appliance domain.	https://ontology.tno.nl/saref.ttl
RealEstate-Core (REC) ontology	The REC ontology provides classes and properties to support the modelling of devices and their features, the modelling of building elements, and the modelling of contractual situations relating to the buildings.	https://doc.realestatecore.io/3.0/full.rdf
Brick Ontology	An ontology that describes the operational aspects of buildings.	https://brickschema.org/schema/Brick

Rule languages, (e.g., Semantic Web Rule Language (SWRL) [56] and Jena [57]) are used for semantic reasoning, which can infer and derive new knowledge from existing information to enrich the RDF graph. Furthermore, a set of predefined vocabularies are provided by the RDF and RDF Schema (RDFS) [58] to define the metadata schema, such that the data representation can be formalised. The above predefined vocabularies can only describe basic relationships between data. Complex knowledge and relationships can be represented by the Web Ontology Language (OWL) [59], which is defined as an ontology that employs a broad set of vocabularies to express domain-specific entities and their relationships.

There have been extensive studies of semantic web-based approaches of data interoperability in the AEC domain. Domain experts have developed a range of ontologies for specific domain information (Table 1). Since ifcOWL maintains the complex structure of the EXPRESS schema leading to low efficiency in practice, simplification based on ifcOWL can be an alternative to domain ontology construction [60,61]. The BOT ontology defined by the W3C linked building data (LBD) group is preferred for smart building DTs [54]. Based on reusing and integrating the above ontologies, linked building data models have been built to manage cross-domain information in building automation and control [62], building energy performance assessment [63], as well as property management [64]. During the above processes, semantic reasoning could facilitate indirect and implicit relationship processing for multi-ontology integration [60,63,65].

Thus, in this study, a BFP ontology is developed as a semantic model that can serve as a bridge between BIM information and sensor data. In this way, the BIM information and the fire scene information are

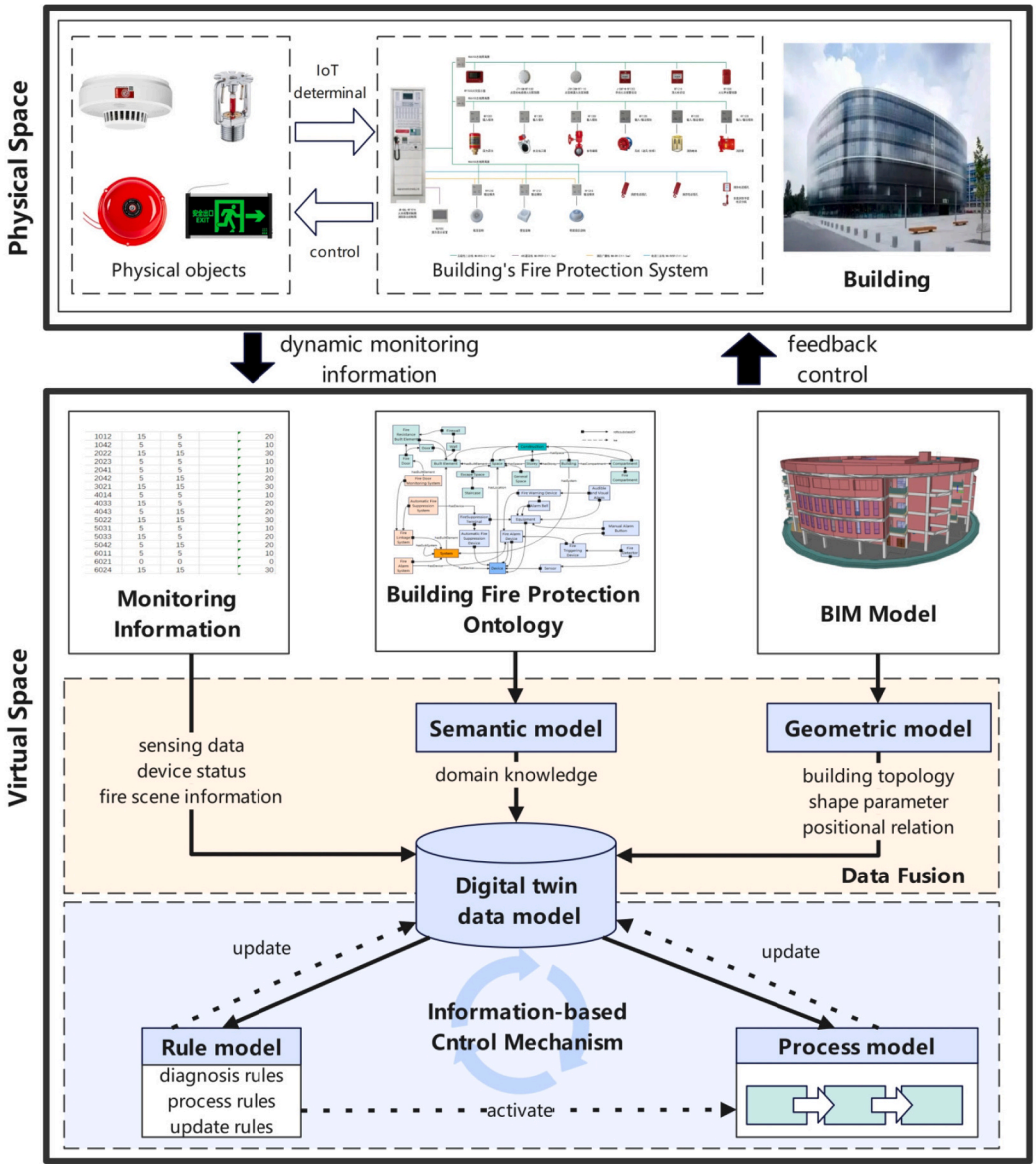


Fig. 1. DT framework for intelligent control of BFP system.

integrated as a linked RDF data model, which is considered as the centre DT data model for DT-driven intelligent control of the fire protection system.

3. Method

3.1. Theoretical framework

Fig. 1 illustrates the framework of the intelligent control of a BFP system. The physical space provides dynamic monitoring information to the virtual space in real-time. In the virtual space, five models (i.e., the geometric model, semantic model, DT data model, rule model, and process model) are established corresponding to the physical space. A deep integration of the above five models is formed as a DT to truly reflect the intelligent control process of a physical BFP system.

The geometric model refers to a 3D model constructed based on BIM and describes static building data of the physical space (e.g., the building topology, shape parameters, and positional relations). The dynamic building data (e.g., the sensing data, device status, and fire scene information) are monitored by different IoT terminals. A BFP ontology is created as the semantic model, which can provide domain knowledge for how the above data are linked, to achieve the fusion of the above two types of data and form a linked-data diagram. The fused data model is called the DT data model and is represented as an RDF graph.

Based on the DT data model, the intelligent control of the BFP system is achieved by a set of information-based control mechanisms that are accomplished by the rule model and process model. The rule model involves several diagnosis rules, process rules, as well as update rules. The process model specifies the control strategies containing several steps and is activated by diagnosis rules. For each step of the control strategy, the process rules are adopted to target the controlling objects based on the DT data model and produce control commands. The above feedbacks are executed in the physical space. Simultaneously, to keep the DT data model always a digital mirror of the physical space, the information in the DT data model (e.g., the device status changes based on the control strategy and real-time monitoring data), is updated according to the update rules. Iteratively, the process model provides a subsequent step of the control strategies in accordance with the updated DT data model, and the control mechanism is achieved.

Table 2
Namespaces and related prefixes of BFP ontology.

Prefix	Description	Namespace
bfp	Concepts of core schema of the BFP ontology.	https://oxazajl.com/BFPOnt/Core#
bfpi	Instances in the DT data model	https://oxazajl.com/BFPOnt/inst#
bot	BOT ontology	https://w3id.org/bot#
brick	Brick ontology	https://brickschema.org/schema/Brick#
sosa	SOSA ontology	http://www.w3.org/ns/sosa/
qudt	Main QUDT ontology	http://qudt.org/schema/qudt/
unit	QUDT units vocabulary.	http://qudt.org/vocab/unit/

3.2. Building fire protection ontology

The BFP ontology is proposed to support the modelling of three information requirements: the geometric topology of the building; the physical devices of the fire protection system and their location, affiliation and technical properties; and the operating information of the fire protection system. During the ontology development procedure, reusing existing ontologies is an important step and can promote the efficiency of development [67]. Accordingly, to meet the above information requirements, the BFP ontology is constructed based on the modular assembly and expansion of existing domain ontologies, such as the BOT ontology, brick ontology, SOSA ontology, and QUDT unit vocabulary. The BOT ontology contains the main topological concepts of buildings, and the above concepts are used to describe the geometric topology of the building in the BFP ontology. Brick ontology provides a standard sematic description of building assets and their relationships. Relational descriptions between devices in the brick ontology are mainly referenced in this study. The SOSA ontology is adopted to express the operating information, including the observed values and time. The units of the observed values are specified using the QUDT unit vocabulary. The namespaces and related prefixes of the ontologies in this study are listed in Table 2. In addition, the above ontologies lack descriptions related to the fire protection domain. The expansion is necessary, and the namespaces of the extensional concepts are defined as "bfp" for distinction. Accordingly, static building information and dynamic building information can be integrated into an RDF graph (i.e., the DT data model) using the terminologies and relationships defined in the BFP ontology. The instances in the DT data model possess the prefix "bfpi".

The BFP Ontology contains four top classes at the highest level of abstraction: *bfp:Construction*, *bfp:System*, *bfp:Device*, and *bfp:Operation*, as presented in Fig. 2. The *bfp:Construction* is the class of building entities, and its core hierarchy structure of subclasses is derived from the BOT ontology. Additionally, more subclasses and relations are specified as follows:

- Two classes are defined as the subclasses of the *bot:Space*, i.e., *bfp:EscapeSpace* and *bfp:GeneralSpace*. The spaces adopted for guiding people to safety in a fire emergency are defined as individuals of the *bfp:EscapeSpace*, such as staircases and corridors. Other spaces are considered individuals of the *bfp:GeneralSpace,* which can also be subdivided into several classes in accordance with their functions (e.g., a reading room and washroom).
- The *bfp:BuiltElement* is defined as a subclass of *bot:Element* and is decomposed based on two parallel criteria. The built elements are classified as doors, walls, windows, floors, and so forth in accordance with usual practice. Additionally, the built elements are classified as fire-resistant built elements and nonfire-resistant built elements based on their technical functions in fire protection. For example, *bfp:FireDoor* can be defined as a subclass of both *bfp:Door* and *bfp:FireResistantBuiltElement*. The correlation between a zone and a built element is defined as *bfp:hasBuiltElement*, which is a sub property of *bot:hasElement*.

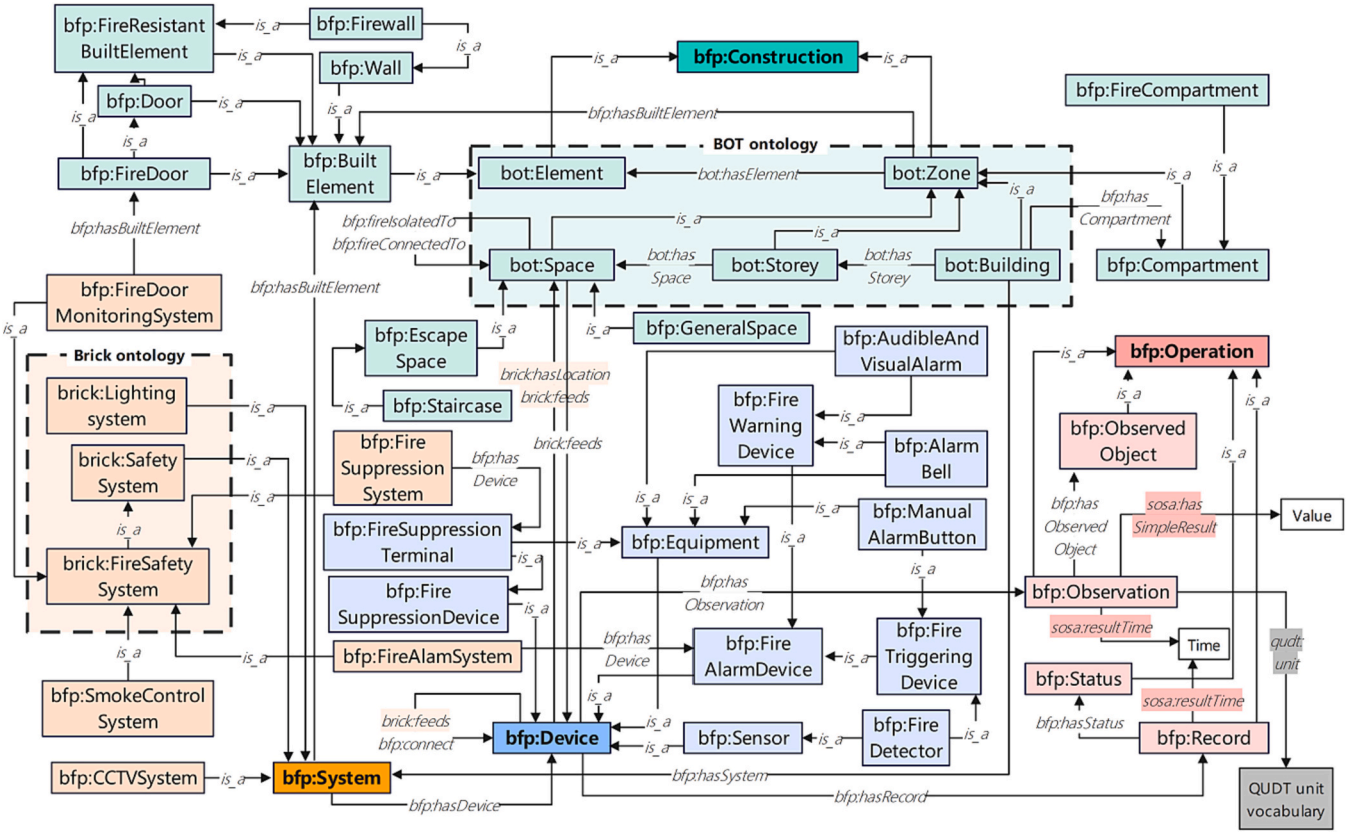


Fig. 2. Illustration of primary classes and relations of BFP ontology.

Table 3
Part of enumerated classes and corresponding value definitions.

Enumerated classes		Example Values
bfp:Status	*bfp:FireAlarmStatus*	*DISABLED, NORMAL, ALARM*
	bfp:SuppressionDeviceStatus	*DISABLED, DEACTIVATE, ACTIVATE*
	bfp:FanStatus	*DISABLED, ON, OFF*
	bfp:InletOutletStatus	*DISABLED, ON, OFF*
	bfp:DoorStatus	*OPEN, CLOSE*
bfp:ObservedObject		*Obscuration, People*

- The *bfp:Compartment* is defined as a subclass of *bot:Zone,* referring to a collection of zones used for a certain purpose. For instance, *bfp: FireCompartment* is a subclass of *bfp:Compartment,* which refers to the area separated by a set of fire-resisting built elements. A fire compartment may contain several spaces and is designed to limit the fire in a certain area, reduce loss, and provide an advantage for escape and rescue. The compartment of a building is defined via *bfp: hasCompartment*.
- Two relations are established between spaces according to their splitting built elements. If two spaces are split by fire resistant elements, the relation between the above two spaces is defined as *bfp: fireIsolatedTo,* which means the fire cannot spread from one space to the other; otherwise, the relation between the two spaces is defined as *bfp:fireConnectedTo*.

A *bfp:System* represents a virtual collection of *bfp:Device* or *bfp:BuiltElement,* which are defined via *bfp:hasDevice* and *bfp:hasBuiltElement*, respectively. The subclasses of *bfp:System* are extended in accordance with the concepts from brick ontology (e.g., a *brick:FireSafetySystem*, a *brick:LightingSystem*, and a *bfp:CCTVSystem*). The *brick:FireSatetySystem* comprise several subclasses, including a *bfp:FireSuppressionSystem* that serves to suppress fire hazards, a *bfp:FireDoorMonitoringSystem* that controls the opening and closing of the fire door, a *bfp:FireAlarmSystem* that alerts people to fire emergencies, a *bfp:SomkeControlSystem* that regulates smoke, and so forth. The relationship indicating that a *bot: Building* possesses a *bfp:System* is defined as *bfp:hasSystem*.

Instances of *bfp:Device* can be classified as equipment or sensors. The *bfp:Equipment* refers to the device that serves a building (e.g., the alarm bell and fire suppression terminal). The *bfp:Sensor* represents measuring devices, including fire detectors (e.g., smoke detectors, heat detectors, and flame detectors). Correlations of the devices are expressed mainly using the relationships defined in brick ontology. The location of a device is specified via *brick:hasLocation*. The sequential relations between devices are expressed through *bfp:connect* or *brick:feeds*. The former places a greater focus more on the connection, whereas the latter stresses the media passing between two devices. Besides, the relation *brick:feeds* can express the media passing from a room to a device. For instance, the relation between a smoke exhaust fan and a room is defined as "(Room, *brick:feeds*, SmokeExhaustFan)", i.e., the smoke is discharged to the outside through the smoke exhaust fan.

The *bfp:Operation* is an abstract class that reports the current building operating situation received from the IoT system, which comprises the working record of the devices (i.e., *bfp:Record*), the status of the devices (i.e., *bfp:Status*), and the observed results of the devices (i.e., *bfp:Observation*), their observed objects (i.e., *bfp: ObservedObject*), and so forth. An individual of the *bfp:Observation* represents an observed result for the devices. The relations *bfp:hasObservation* and *bfp:hasObservedObject* indicate the data source. Likewise, an individual of *bfp:Record* stores the working status via *bfp:hasStatus*. Moreover, relations defined in the SOSA ontology, *sosa:resultTime* and *sosa:hasSimpleResult*, can be adopted to specify the recording time and value. The units of the value are given by the QUDT unit vocabulary via *qudt:unit*. More details are presented in Section 3.3.2. In this study, *bfp:Status* and *bfp:ObservedObject* are defined as enumerated classes with several frequently used values in the operation. Some of the values are listed in Table 3.

Table 4
Examples of mapped terminologies between IFC Schema and BFP Ontology.

IFC Schema	BFP Ontology
IfcBuilding	*bot:Building*
IfcBuildingStorey	*bot:Storey*
IfcSpace	*bot:Space*
IfcDoor	*bfp:Door*
IfcWall	*bfp:Wall*
IfcSystem	*bfp:System*
IfcLightFixture	*bfp:LightFixture*
IfcFireSuppressionTerminal	*bfp:AutomaticFireSuppressionDevice*

3.3. Data fusion

3.3.1. Information extraction from IFC file

The proposed BFP ontology is considered as the backbone of the DT data model. It defines the way in which the static building information and dynamic monitoring information are linked and fused. As mentioned above, the static building information is presented by the geometry model (i.e., BIM model). In practice, the BIM model is created based on different BIM authoring software applications, such that various BIM data formats may be produced. IFC refers to an open data exchange model for BIM application. Since IFC is widely accepted and supported by most BIM authoring software applications, the IFC file serves as the data source for the static building information. Thus, establishing static building information in the DT data model is dependent on information extraction from the IFC file. The extracted IFC entities and their characteristics are defined as instances with attributes in the DT data model in accordance with the mappings between the IFC schema and the BFP ontology. Moreover, the correlations between the IFC entities can be obtained and then defined in the DT data model to generate a linked data graph. In this study, the IFC4 schema is selected for analysis. The information extracted from IFC files mainly involves the description of building topology, the location and aggregation of devices, and the creation of fire compartments. An open-source toolkit, IfcOpenShell [68], is employed for IFC schema parsing.

To describe the building topology in the DT data model, the IFC entities are first defined as the individuals of corresponding classes in accordance with the mapped terminologies between the IFC schema and BFP ontology (Table 4). To specify the subclass to which the IFC entity belongs, predefined properties are required to provide mapping restrictions. For instance, only the instances of *IfcDoor* with a fire rating definition are defined as the individuals of *bfp:FireDoor* (Fig. 3). The predefined fire rating property is stored as an *IfcPropertySingleValue* (#416615), which is connected to *IfcDoor* entity (#416612) via *IfcPropertySet* entity (#416616) and *IfcRelDefinesByProperties* entity (#416621).

Next, the correlations between individuals can be defined in accordance with their related IfcRel-entities (Table 5). SPARQL queries can be employed to supplement the relations. For example, the query " *SELECT ?a ?b WHERE{?a rdf:type bfp:SmokeExhaustOutlet. ?b rdf:type bfp: SmokeExhaustFan. ?a bfp:connect+ ?b .}*" can be adopted to retrieve several pairs of smoke exhaust fans and linking outlets. Subsequently, the *brick:feeds* relation is created, starting from the smoke exhaust outlet to the connected smoke exhaust fan.

The device locations (i.e., *brick:hasLocation*) are created by geometric processing of the IFC file. Since IfcOpenShell can convert the implicit geometry in IFC files into explicit geometry, the Boolean intersection operation is executed for the respective pair of devices and spaces. If the interaction set is not empty, the space is defined as the location of the device. In addition, semantic reasoning can be adopted to obtain the aggregation relationships between the system and devices. For instance, the instances of *bfp:FireAlarmSystem* refer to the systems with several fire alarm devices and thus can be modelled by the reasoning in accordance with the semantic rule "(*?a bfp:hasDevice ?b*)(*?b rdf:type bfp:FireAlarmDevice*)->(*?a rdf:type bfp:FireAlarmSystem*)".

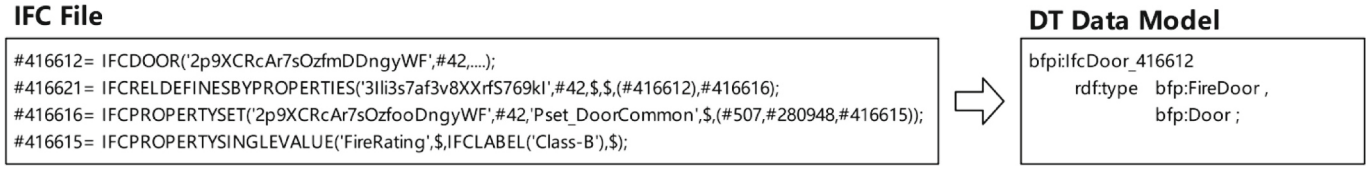


Fig. 3. Using predefined properties to map information in IFC with BFP ontology.

Table 5
Relation definitions in DT data model according to related IfcRel-entities.

IFC File	DT Data Model
#209 = IFCBUILDINGSTOREY(…); #142 = IFCBUILDING(…); #1417063 = IFCRELAGGREGATES('…',#42, $,$,#142,(…,#209));	bfpi:IfcBuilding_142 *bot:hasStorey* bfpi:IfcBuildingStorey_209
#209 = IFCBUILDINGSTOREY(…); #35344 = IFCSPACE(…); #1417019 = IFCRELAGGREGATES('…',#42, $,$,#209,(#35344, …));	bfpi:IfcBuildingStorey_209 *bot:hasSpace* bfpi:IfcSapce_35344
#148573 = IFCWALL(…); #1533 = IFCSPACE(…); #1435550 = IFCRELSPACEBOUNDARY('…', #42,'1stLevel',$,#1533, #148573,#1695,.PHYSICAL.,.EXTERNAL.);	bfpi:IfcSpace_1533 *bfp:hasBuiltElement* bfpi:IfcWall_148573
#854413 = IFCALARM(…) #1090998 = IFCSYSTEM(…); #1091002 = IFCRELASSIGNSTOGROUP('…', #42,$,$,(#854413, …), .PRODUCT.,#1090998);	bfpi:IfcSystem_1090998 *bfp:hasDevice* bfpi:IfcAlarm_854413
#142 = IFCBUILDING(…); #1090998 = IFCSYSTEM(…); #1090999 = IFCRELSERVICESBUILDINGS ('…',#42,$,$,#1090998,(#141));	bfpi:IfcBuilding_142 *bfp:hasSystem* bfpi:IfcSystem_1090998
#5886 = IFCBUILDINGELEMENTPROXY(…); #8322 = IFCDUCTFITTING(…); #9765 = IFCRELNESTS('…', #42,'NestedPorts','Flow',#5886,(…,#9564)); #9564 = IFCDISTRIBUTIONPORT(…); #9573 = IFCRELCONNECTSPORTS('…', #42,'…','Flow',#9564,#9571,$); #9571 = IFCDISTRIBUTIONPORT(…); #9776 = IFCRELNESTS('…','Flow',#8322, (#9571,#9620));	bfpi: IfcBuildingElementProxy_5886 *bfp:connect* bfpi:IfcDuctFitting_8322

Since the IFC lacks a description of some domain-specific concepts, the information of the fire compartment cannot be directly acquired from IFC. As a result, starting from the definition of the fire compartment that represents a limited range of areas separated by fireproofing constructions, the disjoint-set union-finding algorithm is used to create fire compartments, as presented in Fig. 4. Two relations (i.e., *bfp:fireIsolatedTo* and *bfp:fireConnectedTo*) are created between spaces according to their splitting built elements first. For example, the relation between Space_1 and Space_2 is expressed as *bfp:fireConnectedTo* for their relating non-fire door (i.e., Door_3), while the relation between Space_1 and Space_3 is expressed as *bfp:fireIsolatedTo* since they are split by a fire door (i.e., Door_2). Next, a set of connected graphs is obtained based on the above two relations. The nodes of the graphs represent the spaces, and the edges are created based on the *bfp:fireConnectedTo* relations. Notably, those connected spaces belong to the same fire compartment. As a result, the creation of fire compartments is formalised as a union-finding problem. In this way, the spaces are organised as a disjoint-set data structure, and the respective disjoint-set is defined as an instance of *bfp:fireCompartment*. The spaces are connected to the fire compartment via *bot:hasSpace*.

Finally, the completeness and correctness of the DT data model are important basics of the implementation of the use of the two technologies for the intelligent control of BFP systems in fire accidents. A set of verification rules are developed based on SPARQL to ensure an accurate transformation from the IFC file to the DT data model. The verification rules can be divided into three types, and some examples are listed in Table 6. First, the SPARQL query in *SELECT* form with *COUNT* clauses is employed for a consistency check between the IFC file and DT data model. For instance, the verification rule VR_1 is used to obtain the number of spaces in the DT data model, which is then compared with the number of spaces in the IFC file. Second, the classification check rules, which are used to ensure accurate classification of individuals according to the BFP ontology, are developed in *ASK* form and result in Boolean values (i.e., "True" or "False", which means the DT data model passes or fails in the verification process, respectively). For instance, the verification rule VR_2 is adopted to check whether the classification of spaces is correct. According to the BFP ontology, the number of individuals of *bot:Space* (i.e., the variable *?all* in VR_2) should be equal to the sum of those of *bfp:EscapeSpace* and *bfp:GeneralSpace* (i.e., the variable *?s1* and *?s2* in VR_2). The comparison is achieved by the *HAVING* clause. The third type of verification rule is also developed in *ASK* form and is used to ensure accurate compositional relationships between individuals. For example, a CCTV system is composed of at least one camera, and this compositional relationship is verified using the verification rule VR_3 with the *FILTER NOT EXISTS* clause.

3.3.2. Description of IoT information in DT data model

In the DT data model, the information collected from IoT devices is described as individuals of *bfp:Observation* or *bfp:Record*. The observed time and value are specified by *sosa:hasSimpleResult* and *sosa:resultTime*, respectively. The unit of observation is indicated via *qudt:unit*. As depicted in Fig. 5, the observed obscuration of the smoke detector, IfcAlarm_930815, is 0%/m, and its status is "*NORMAL*".

3.4. Information-based control mechanism

The control mechanism of the DT framework is established based on the rule model and the process model, as presented in Fig. 6. The rule model contains a set of diagnosis rules, process rules, and update rules. The diagnosis rules are executed on the time slice of the DT data model at set intervals to check if a fire is detected. Time slices of the DT data model can be obtained by SPARQL queries. For instance, the SPARQL query in Fig. 6 is employed to construct a sub data model from 13:38:00 to 13:38:10. At each time interval, a diagnosis rule is executed to check for the fire incidents. If none is found, the diagnosis rule is executed on the DT data model of the subsequent time interval. Otherwise, the process model is activated. Process rules are adopted to complete the tasks within the process model. Meanwhile, update rules are employed to update the status of entities and the information collected from IoT devices into the DT data model.

The process model is developed based on BPMN and executed using the Camunda Platform as a workflow engine. A typical process is modelled as presented in Fig. 7. As mentioned before, the intelligent control strategy consists of several subtasks that are completed based on the process rules; model updates are implemented based on update rules. The above two processes constitute a single cycle. A judgement condition is also set to determine whether the loop is terminated, which can be either an automatic judgement criterion or a manual judgement

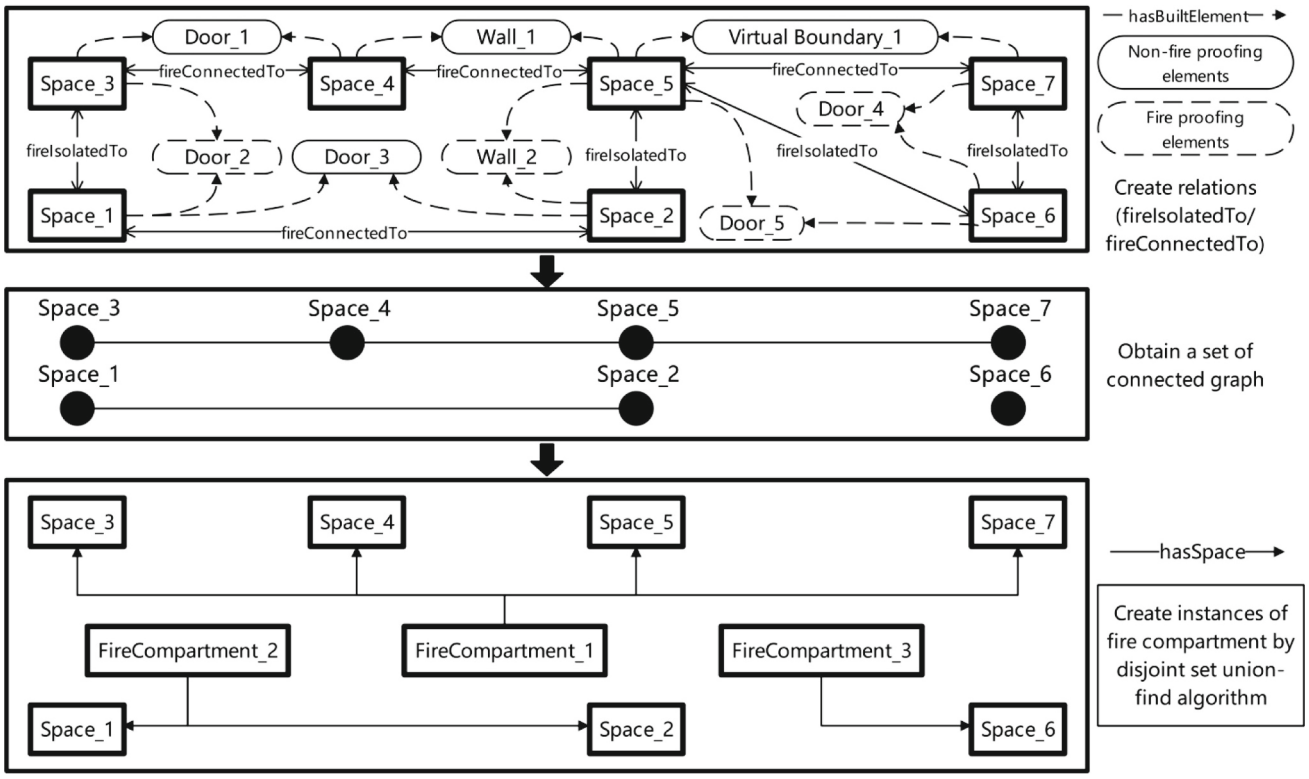


Fig. 4. Approach to create fire compartment.

Table 6
Examples of verification rules.

No.	Type	Example of verification rule	Description
VR_1	Consistency check rule	*SELECT (COUNT(?obj) AS ?count) WHERE{ ?obj rdf:type bot:Space .}*	To obtain the number of individuals of spaces in DT data model.
VR_2	Classification check rule	*ASK WHERE { {SELECT ?all ?s1 ?s2 WHERE { {SELECT (COUNT(?obj) AS ?all) WHERE {?obj rdf:type bot:Space}} {SELECT (COUNT(?obj) AS ?s1) WHERE {?obj rdf:type bfp:EscapeSpace}} {SELECT (COUNT(?obj) AS ?s2) WHERE {?obj rdf:type bfp:GeneralSpace}} } HAVING (?all = (?s1 +?s2)) } }*	To verify the classification of spaces.
VR_3	Composition check rule	*ASK WHERE { {SELECT ?sys WHERE {?sys rdf:type brick: CCTVSystem . ?dev rdf:type bfp: Camera . FILTER NOT EXISTS { ?sys bfp:hasDevice?dev} } } }*	To verify the composition of CCTV system.

```
bfpi:IfcAlarm_930815  rdf:type  bfp:SmokeDetector;
    brick:hasLocation      bfpi:IfcSpace_24833 ;
    bfp:hasObservation   bfpi:Observation_19794 ;
    bfp:hasRecord        bfpi:Record_19794 .

bfpi:Observation_19794
    rdf:type                bfp:Operation , bfp:Observation ;
    bfp:hasObservedObject  bfp:Obscuration ;
    qudt:unit               unit:PER-M ;
    sosa:hasSimpleResult    "0.0"^^xsd:float ;
    sosa:resultTime    "2022-05-02T21:36:13.901615+00:00"^^xsd:dateTime .

bfpi:Record_19794  rdf:type  bfp:Operation , bfp:Record ;
    bfp:hasStatus     bfp:NORMAL ;
    sosa:resultTime    "2022-05-02T21:36:13.901615+00:00"^^xsd:dateTime .
```

Fig. 5. Fragment of DT data model describing IoT information.

criterion. In the next cycle, the intelligent control strategy is created based on the updated DT data model. It is important to note that rule models and process models are different in various AEC projects, and their creation depends on domain experts. For different fire emergencies and to achieve different control strategies, domain experts can create or modify the rules and the processes based on their experience.

4. Case study

4.1. Preparation

In this study, a 4-storey school library was adopted to verify the use of DTs and the semantic web technologies for the intelligent control of BFP systems in fire accidents. The BIM model created based on Autodesk

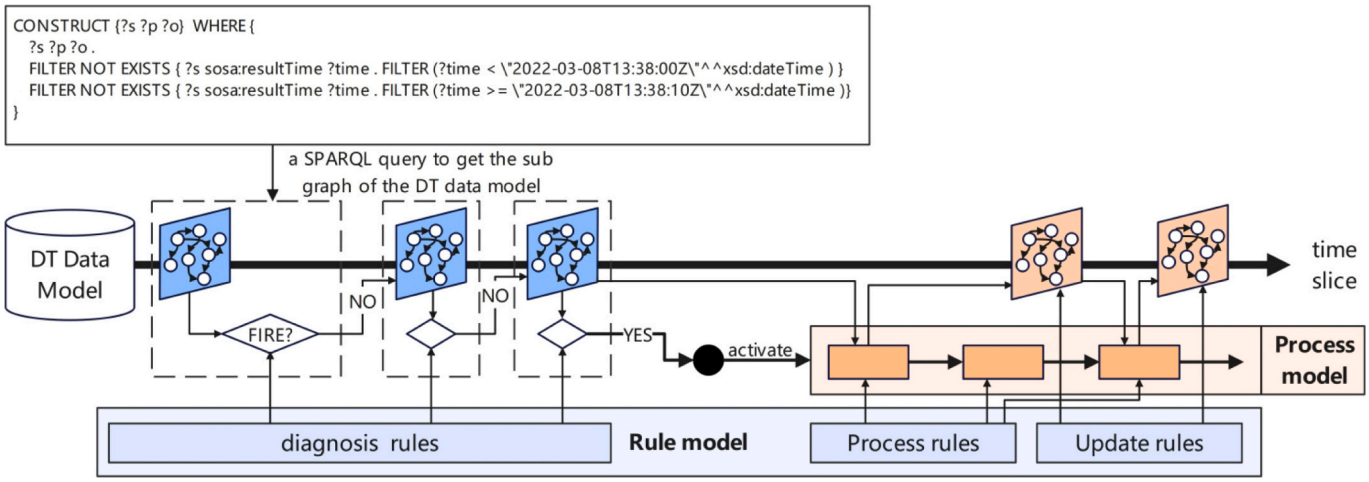


Fig. 6. Information-based control mechanism.

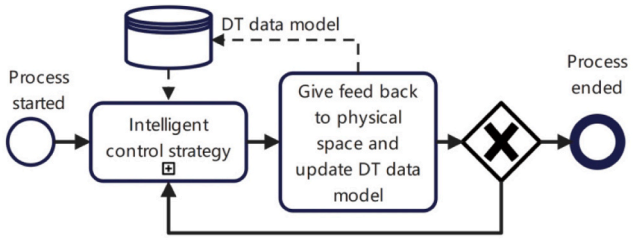


Fig. 7. Typical process model.

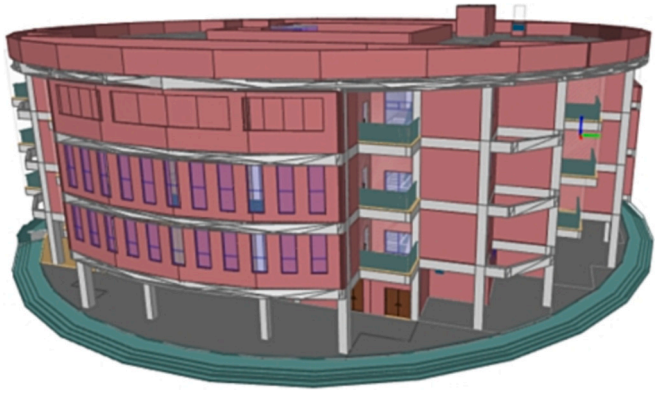


Fig. 8. Designed BIM model.

Revit 2022, as presented in Fig. 8, was employed as the geometric model. The BIM model was exported as an IFC file using the IFC4 design transfer view with a first-level space boundary.

Since on-site experiments are difficult to conduct, computer simulation, which can also effectively reflect real situations, was conducted in this study to verify the feasibility of the use of the two technologies. The computational fluid dynamics (CFD) simulation results based on the fire dynamics simulator (FDS) and related FDS tools (e.g., Pyrosim and Smokeview) were the dynamic monitoring data collected by the IoT sensors. The relative simulation parameters were defined in accordance with [69,70]. The environmental temperature was 26 °C, and the relative humidity was 40%. The grid size was set as 0.8 m × 0.8 m × 0.8 m. A circular fire with the t^2 growth of the heat release rate was located on the centre of the room with a combustible surface of 2 × 4 m. The fire power was defined as $Q = 8000$ kW. On that basis, the ramp-up time $t = 413$ s was calculated by $Q = \alpha t^2$, where the growth factor α was set as 0.04689. The fire ignited at the central point of the combustible surface and propagated outwards at $V_0 = 0.01$ m/s. The extinguishing coefficient of the burner surface was set as 0.1 (m^2/kg)/s. It is assumed that the system works starting at 13:38:00 and the diagnosis rules are executed every 10 s. Furthermore, the simulation was used to examine the performance of the proposed approach.

The evacuation simulation results based on Pathfinder were the real-time monitoring data captured by cameras. The densities of the rooms varied with the functions; for example, the density of the reading room was 4.65 m^2 per person, and the density of the meeting room was 0.65 m^2 per person. Thus, 922 occupants were added to the building in total. The added occupants were set as young people, and the male to female ratio was set at 1:1. Moreover, the pre-evacuation time of the people in different rooms varied with their distance to the fire origination. Given the mental factors, people who are closer to where the fire begins will have a shorter pre-evacuation time. Thus, based on the SFPE Handbook of Fire Protection Engineering, the pre-evacuation time of the people in the room of fire origination was set as 90 s, and the pre-evacuation time of the people in the room located on the same floor as the fire origination was set as 150 s. Besides, the rest was set as 210 s.

4.2. DT data model construction

A program based on the algorithms and methods mentioned in Section 3.3 was implemented for DT data model construction. The program was deployed on the Java platform, and Jena was adopted to build semantic web applications. As mentioned earlier, IfcOpenShell was adopted to extract static building information from the IFC file. This extracted information was stored as several CSV files, as presented in Fig. 9. Instances were created in the DT data model by processing the CSV files line-by-line. Dynamic monitoring information was created from the IoT database. The mappings between the GUID of IFC entities and the ID of recording objects in the IoT database were employed to achieve information integration. Moreover, in the above process, an object detection model (e.g., YOLOV4 [72]) was used to determine the number of people in a room based on the real-time pictures or videos collected by the cameras [73], as presented in Fig. 10. The identified results were stored in the IoT database and then instantiated in the DT data model.

Fig. 11 illustrates part of the description of the DT data model. The initial status of all the sprinklers was set as "*DEACTIVATE*". Fire doors were normally closed but kept open during evacuation to facilitate the flow of people. Given the most unfavourable scenario, the initial status of the doors was set as "*OPEN*" during the simulation. Since the

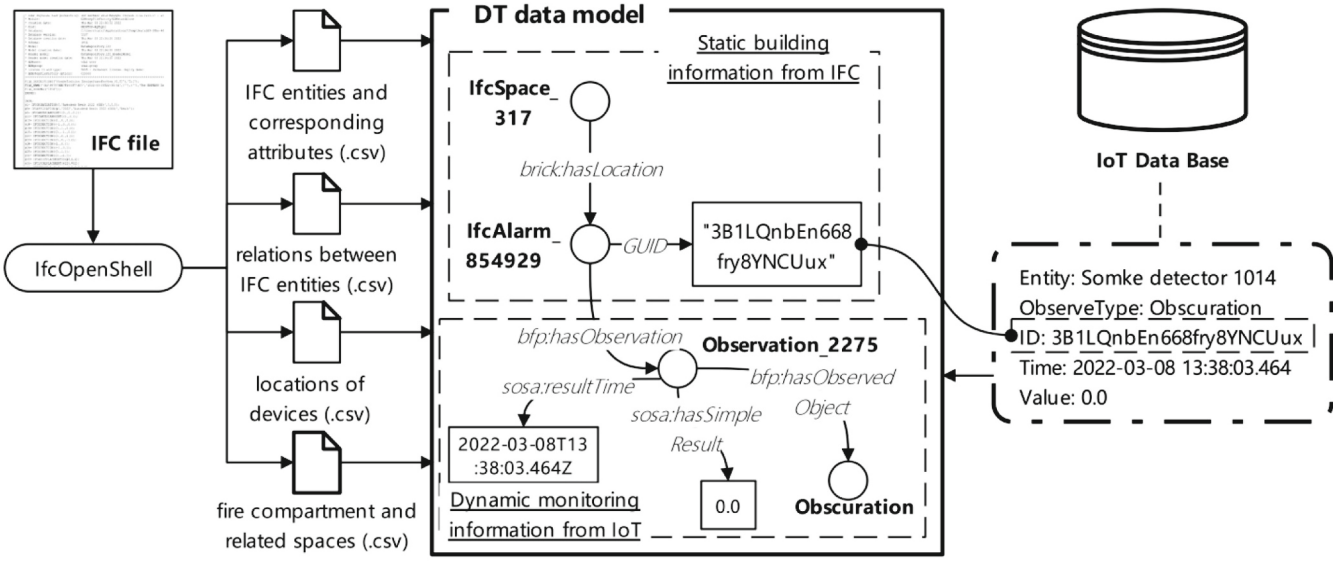


Fig. 9. Implementation framework of DT data model construction.

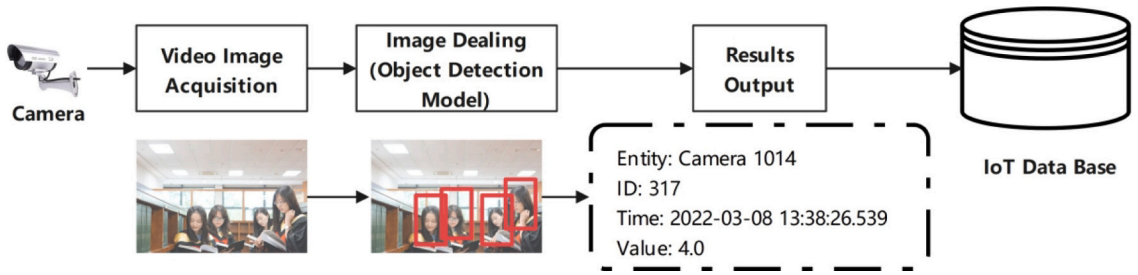


Fig. 10. Using object detection model to count people for each space.

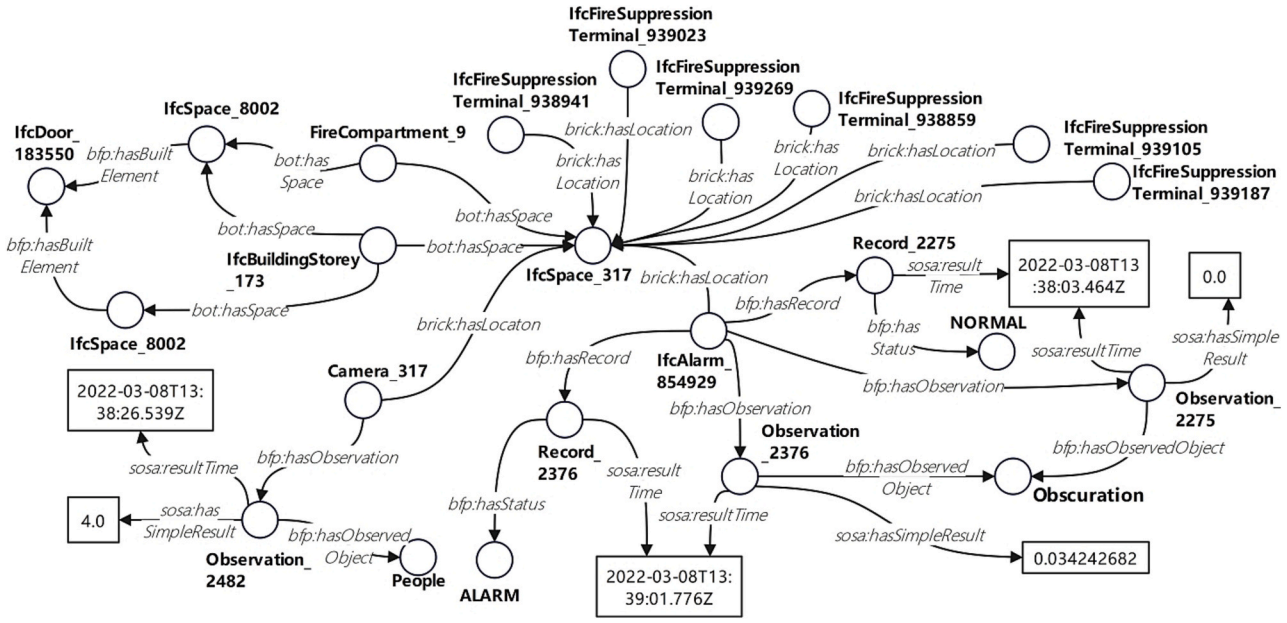


Fig. 11. Part of DT data model.

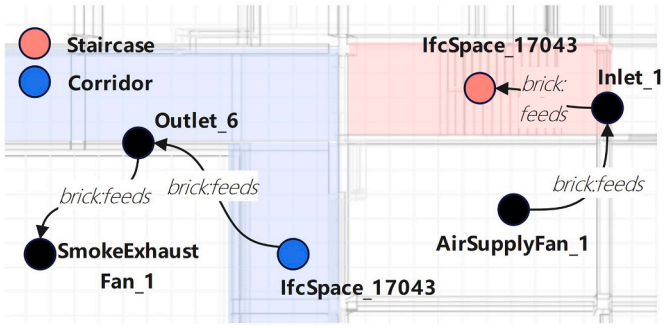


Fig. 12. Flow description in DT data model.

obscuration threshold of the smoke detector was 0.0328/m, the smoke detector *IfcAlarm_854249* alarmed at 13:39:02. Furthermore, the information captured by the cameras was described in the RDF data model, such as the node *Camera_317* in Fig. 11.

The flow description in the DT data model is illustrated in Fig. 12. The direction of the flow is depicted by the *brick:feeds* relation. In the designed building, the respective staircase was equipped with an air supply inlet connected to the air supply fan, and the corridor was equipped with several smoke exhaust outlets connected to the smoke exhaust fan. The initial status of devices in the smoke control system was set as "*OFF*".

4.3. Control mechanism achievement in different scenarios

4.3.1. Intelligent control of sprinklers

The intelligent control mechanism of sprinklers follows the idea of group activation [50]. Once a smoke detector alarms, the sprinklers located in the same room around the smoke detector are activated. The process model is presented in Fig. 13. Rules that are listed in Table 7 were set as the basic rules of the rule model. The diagnosis rule, R1, whose result was a Boolean, checked whether there was an alarming smoke sensor. If the Boolean is "*true*", the process rule R2 is adopted to retrieve the sprinklers located in the same room. The activation signals were sent to the selected sprinklers. Meantime, the update rule, R3, was adopted to update the status of the selected sprinklers in the DT data model.

A 45 m^2 room located on the first floor was selected to simulate the intelligent control mechanism. Since the smoke detector alarmed at 13:39:02, given the calculation and transmit delay, the system detected the smoke detector alarm at 13:39:10 and activated the corresponding sprinklers at 13:39:13. Moreover, the conventional thermal activation mechanism was simulated for comparison. For thermal activation, all the sprinklers were not activated till the bulb temperature reached 68 °C. The response time index of the sprinklers was set as 148 $(m{\cdot}s)^{1/2}$. The parameters of one single sprinkler for both activation mechanisms are listed in Table 8.

Fig. 14 compares the two activation mechanisms. Under the thermal control mechanism, the sprinklers were activated approximately 152 s

Table 7
Basic rules of rule model for intelligent control of sprinklers.

No.	Rule type	Representation form	Rule
R1	Diagnosis rule	SPARQL	*ASK {* *?alarm rdf:type bfp:SmokeDetecor .* *?alarm bfp:hasRecord?record .* *?record bfp:hasStatus bfp:ALARM .* *}*
R2	Process rule	SPARQL	*SELECT?sp WHERE {* *?alarm rdf:type bfp: SmokeDetecor .* *?alarm bfp:hasRecord?record .* *?record bfp:hasStatus bfp:ALARM .* *?alarm brick:hasLocation?space .* *?sp rdf:type bfp:Sprinkler .* *?sp brick:hasLocation?space .* *}*

R3	Update rule	Pseudo code	**Input**: selected sprinklers *sps*; RDF data model *m*; updated time *t*
			Output: Updated RDF data model *m*
			for*sp***in***sps***do**
			// add *r* as an individual of *bfp:Record* in *m*
			m.addIndividual(*r*, *bfp:Record*);
			m.add(*r*, *bfp:hasStatus*, *bfp:ACTIVATE*);
			m.add(*r*, *sosa:resultTime*, *t*);
			m.add(*sp*, *bfp:hasRecord*, *r*);
			end for

Table 8
Parameters of one single sprinkler defined for simulation.

Parameters	Value
Water flow rate	57.6 L/min
Initial droplet velocity	15 m/s
Inner cone angle	45°
Outer cone angle	60°

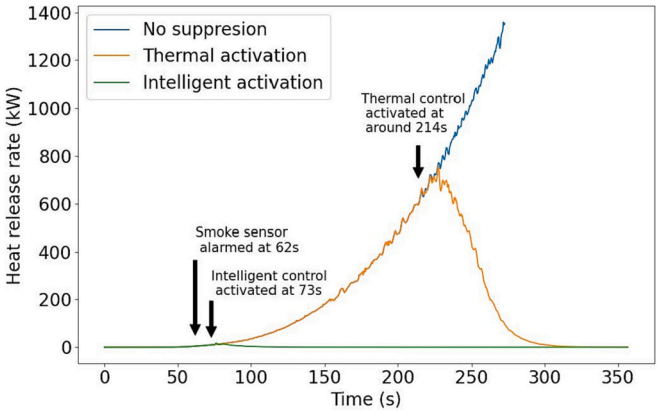


Fig. 14. Dependence of heat release rate on time for different activation mechanisms.

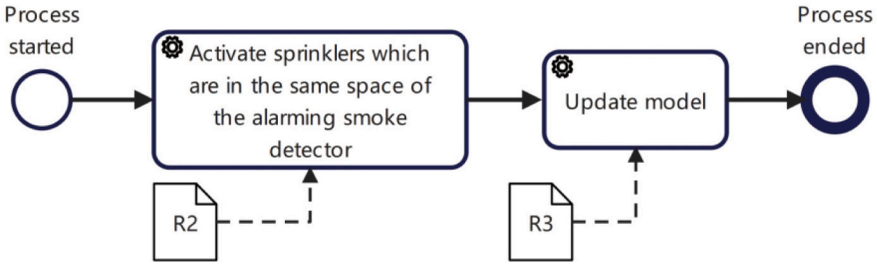


Fig. 13. Process model for intelligent control of sprinklers.

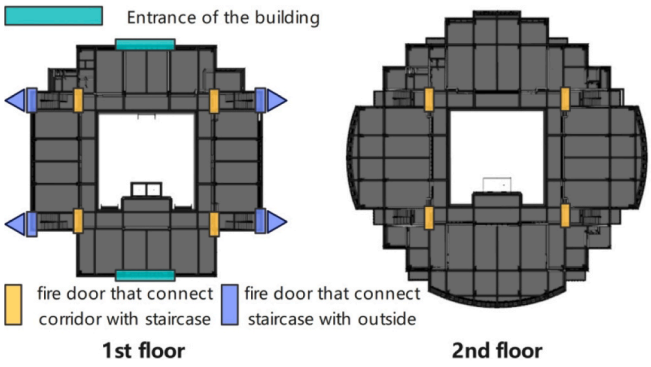


Fig. 15. Floor plans of library.

after the smoke sensor alarm. The fire was suppressed after 320 s. In contrast, the intelligent control mechanism is capable of suppressing fire at a very early stage.

4.3.2. Intelligent control of fire doors

Smoke is another dangerous lethal factor for a fire accident, especially during evacuation. One of the functions of fire doors is to stop the fire and smoke spreading from one fire compartment to another. However, a fire door related to the staircase also serves for evacuation. The time to turn off fire doors that can enable people to escape and prevent broadening the spread of smoke should be determined.

The floor plans of the library are illustrated in Fig. 15. In this study, two strategies were considered in controlling fire doors. First, the fire doors located on the first floor were turned off immediately, and people on the first floor were guided to escape through the two main entrances. Second, the fire doors located on the other floors were turned off after all the people escaped from this floor. The camera was beneficial to capture the evacuation progress.

The process model established for this scenario is shown in Fig. 16. The rule R1, as introduced in Table 8, was also considered as the diagnosis rule in this scenario. The process rules and update rules are listed in Table 9. The SPARL query outlined as R4 was employed to obtain the fire doors located on the first floor. The above fire doors were closed as soon as the smoke detector was alarming. Next, a process rule, R5, can be adopted to examine the number of existing people on the respective floor except for the first floor. A *VALUE* restriction specifies which floor was examined. The calculation was performed based on the cameras' observation. The filter restriction "*FILTER NOT EXISTS {?space rdf:type bfp:Staircase.}*" indicated that the number of people in the staircase was not included in the calculation. The sequential restriction "*ORDER BY DESC(?time) LIMIT 1*" ensured that the query results of the inner *SELECT* clauses were sorted by the reversed order of examining time, and only the first result was considered. Accordingly, the examination was performed on the latest observations. If the result of R5 is "*true*", the process rule, R6, can be used to retrieve the corresponding fire doors. The operation signals were transmitted to the above fire doors, and their status was updated in the DT data model using update rule R9 simultaneously. Afterwards, the process model checked whether all the fire doors were closed based on the process rules R7 and R8. The process rule R7 was used to obtain all the fire doors connecting the staircase and corridor, and the process rule R8 was adopted to check the status of all fire doors. If there are still fire doors that are not closed, the camera monitoring data can be updated into the DT data model by the update rule R10. The process model executed the above tasks iteratively till all the fire doors were closed.

In this scenario, a room located on the second floor was selected as the fire origination to verify the proposed process model. The smoke detector alarmed at 13:38:27, and the closing time of the fire doors is presented in Fig. 17. Another strategy in which the fire doors remained open to ensure evacuation was also simulated for comparison. The comparison results represented in Fig. 18 suggested that the intelligent control mechanism can keep most smoke outside the staircases and ensure evacuation simultaneously.

4.3.3. Intelligent control of smoke control systems

The smoke control system also plays a critical role in preventing damage caused by smoke. In accordance with the Code for Design of Automatic Fire Alarm System in China, the smoke control system is required to start up in the fire compartment where there are two independently alarming smoke detectors. An earlier start-up time can improve the performance of smoke control [71]. However, the start-up time of the smoke control system heavily relies on manual activation, including estimating the fire location and confirming the present fire compartment. In some cases, people should find the corresponding smoke exhaust fan and turn on it by hand, which is time-consuming.

In this study, a process model (Fig. 19) and a set of rules (Table. 10) are proposed to achieve the intelligent control mechanism of the smoke control system. First, the diagnosis rule, R11, was employed to check whether there were two alarming smoke detectors for the respective fire compartment. The process model was activated when the query result of rule R11 was "*true*". Subsequently, two strategies were considered in the process model. First, the exhaust fans of the fire compartment where there were two independent alarming smoke detectors started. The process rule, R12, was adopted to find the corresponding outlets and fans. The inner *SELECT* clauses aim at limiting the variable *?fc* as the target fire compartment. Second, the air supply fan of the staircase through which people were escaping became open. For each staircase, the process rule R13 was used to examine the number of existing people. If the query result of R13 is "*true*", the process rule, R14, is used to obtain the corresponding devices. For both strategies, the opening signals were transmitted to the devices, and the model was updated based on the update rule R16 simultaneously. Next, rule R15 was performed for the respective device of the air supply system to check if it was working sufficiently. If not, the DT data model was updated according to R10, as introduced in Table 9, and the process model executed the next iteration.

The controlling strategy given by the proposed intelligent mechanism is displayed in Fig. 20. The first alarming smoke detector was

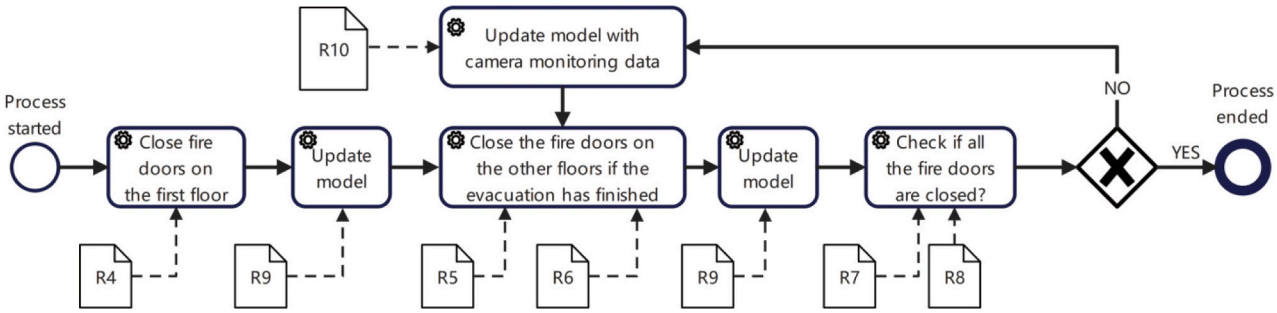


Fig. 16. Process model for intelligent control of fire doors.

Table 9
Basic rules of rule model for intelligent control of fire doors.

No.	Rule type	Representation form	Rule
R4	Process rule	SPARQL	*SELECT?door WHERE {* *?st rdf:type bfp:FirstFloor .* *?st bot:hasSpace?staircase .* *?st bot:hasSpace?corridor .* *?staircase rdf:type bfp:Staircase .* *?staircase bfp:hasBuiltElement?door .* *?door rdf:type bfp:FireDoor .* *?corridor rdf:type bfp:Corridor .* *?corridor bfp:hasBuiltElement?door .* *}*
R5	Process rule	SPARQL	*ASK {* *{SELECT?floor WHERE {* *VALUES?floor {bfpi:* *IfcBuildingStorey_185}* *?floor bot:hasSpace?space .* *?camera rdf:type bfp:Camera .* *?camera brick:hasLocation?space .* *?camera bfp:hasObservation?Observation .* *?Observation sosa:hasSimpleResult?people .* *FILTER NOT EXISTS {?floor rdf:type bfp:FirstFloor .}* *FILTER NOT EXISTS {?space rdf:type bfp:Staircase .}* *?Observation sosa:resultTime?time .* *} GROUP BY?time?floor HAVING (SUM(?people) = 0)* *ORDER BY DESC(?time) LIMIT 1* *}* *}*
R6	Process rule	SPARQL	*SELECT?door WHERE {* *?floor bot:hasSpace?staircase .* *?floor bot:hasSpace?corridor .* *?staircase rdf:type bfp:Staircase .* *?staircase bfp:hasBuiltElement?door .* *?door rdf:type bfp:FireDoor .* *?corridor rdf:type bfp:Corridor .* *?corridor bfp:hasBuiltElement?door .* *VALUES?floor {bfpi:* *IfcBuildingStorey_185}* *}*
R7	Process rule	SPARQL	*SELECT DISTINCT?door WHERE {* *?staircase rdf:type bfp:Staircase .* *?staircase bfp:hasBuiltElement?door .* *?door rdf:type bfp:FireDoor .* *?corridor rdf:type bfp:Corridor .* *?corridor bfp:hasBuiltElement?door .* *}*
R8	Process rule	SPARQL	*ASK {* *?door bfp:hasRecord?record .* *VALUES?door {bfpi:IfcDoor_183550}* *{SELECT?time?status WHERE {* *?record sosa:resultTime?time .}* *ORDER BY DESC(?time) LIMIT 1}* *?record bfp:hasStatus bfp:CLOSE .* *}*

R9	Update rule	Pseudo code	**Input**: selected doors *ds*; RDF data model *m*; updated time *t*
			Output: Updated RDF data model *m*
			for *d* **in** *ds* **do**
			// add *r* as an individual of *bfp:Record* in *m*
			m.addIndividual(*r*, *bfp:Record*);
			m.add(*r*, *bfp:hasStatus*, *bfp:CLOSE*);
			m.add(*r*, *sosa:resultTime*, *t*);
			m.add(*sp*, *bfp:hasRecord*, *r*);
			end for

R10	Update rule	Pseudo code	**Input**: camera data *data*; RDF data model *m*; updated time *t*
			Output: Updated RDF data model *m*
			for (*camera, d)* **in** *data* **do**
			// add *o* as an individual of *bfp:Observation* in *m*
			m.addIndividual(*o*, *bfp:Observation*);
			m.add(*o*, *sosa:hasSimpleResult*, *d*);
			m.add(*o*, *sosa:resultTime*, *t*);
			m.add(*o*, *bfp:hasObservedObject*, *bfp:People*);
			m.add(*camera*, *bfp:hasObservation*, *o*);
			end for

Table 9 (*continued*)

No.	Rule type	Representation form	Rule

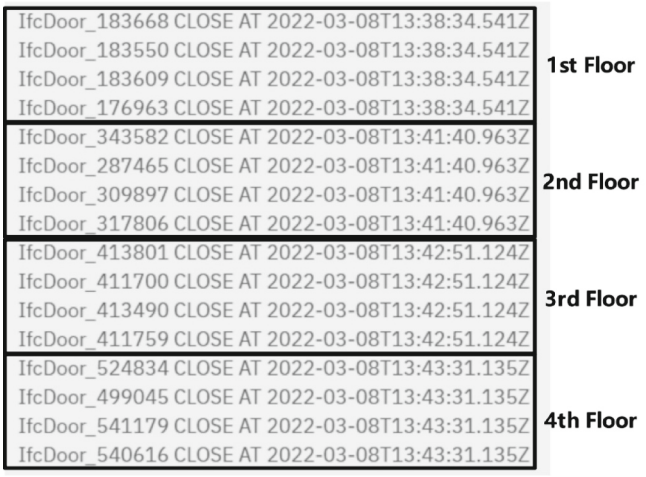


Fig. 17. Closing time of fire doors on different floors.

activated at 13:38:27, and the second alarming smoke detector was activated at 13:39:09. Taking the transmit delay and start-up delay into consideration, the devices began to work 15 s after receiving the opening signals. The air supply fans and related inlets of the staircase on the left side started earlier than those on the opposite side since the people in the fire origination area started to escape earlier than those in the other rooms.

In addition, another strategy considering the manual delay was simulated for comparison. Due to the manual delay, which was assumed as 5 min, the smoke control system started to work at 13:44:25. As depicted in Fig. 21, the smoke was controlled within a limited space under the intelligent control mechanism of the smoke control system and spread more slowly. Moreover, the smoke was kept mostly outside the staircase due to the air pressure difference caused by the continual air supply.

5. Discussion

DTs have been considered a comprehensive solution for improving the efficiency of building automation systems. The core of DTs is the DT model, which enables organic association of multiple pieces of information and supports virtual-real interaction and optimal control on this basis. Focusing on data integration and system cooperation of the intelligent fire protection system, this study proposes a DT-based approach of the intelligent control of building automation systems. From the perspective of information flow, the use of the two technologies for the intelligent control of BFP systems in fire accidents contains a data fusion stage and several information-based control mechanisms. Data fusion aims to use a unified data model to organise and store the information from various data sources. The information-based control mechanisms serve as the interaction between the real and the virtual. From the perspective of modelling, the use of the two technologies consists of five basic models, namely, the geometric model, semantic model, DT data model, rule model, and process model. BIM models are

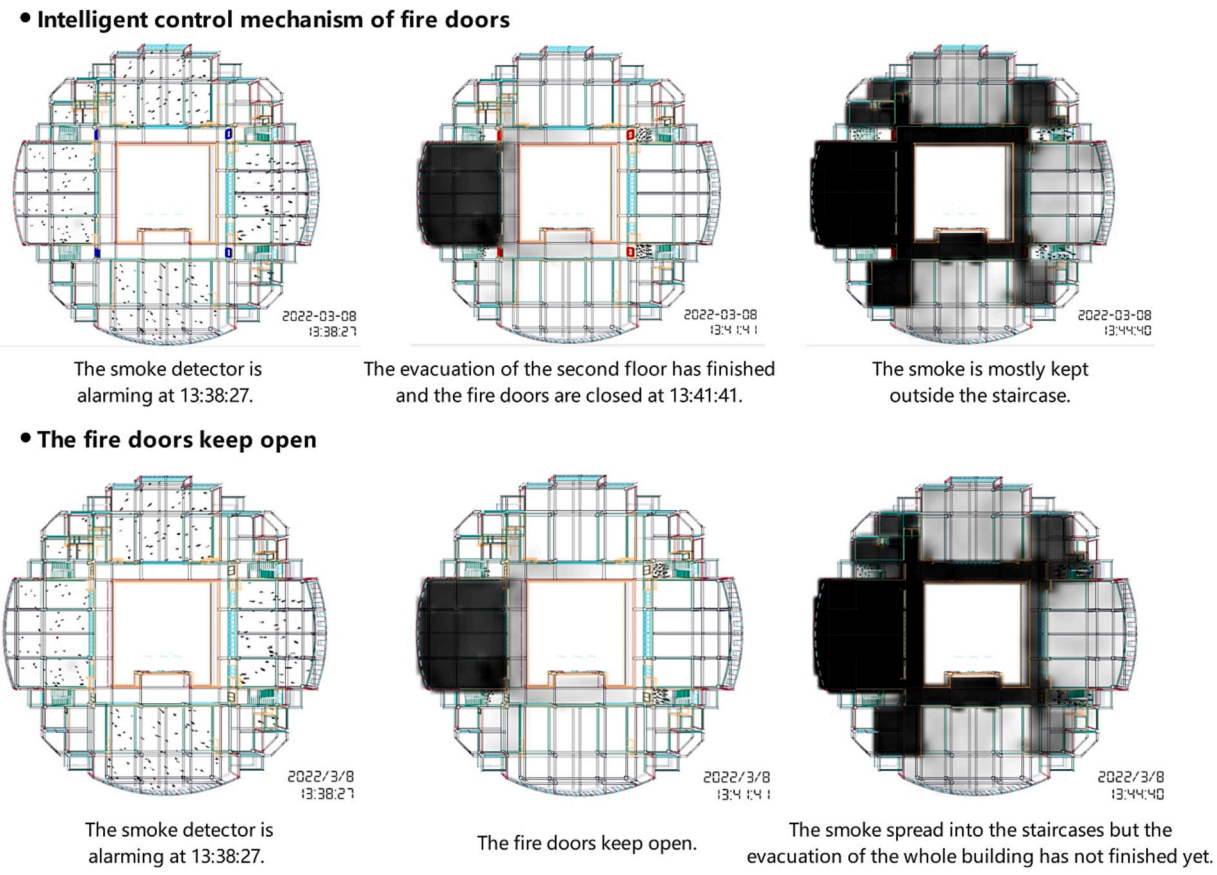


Fig. 18. Smoke distribution under different control mechanisms of fire doors.

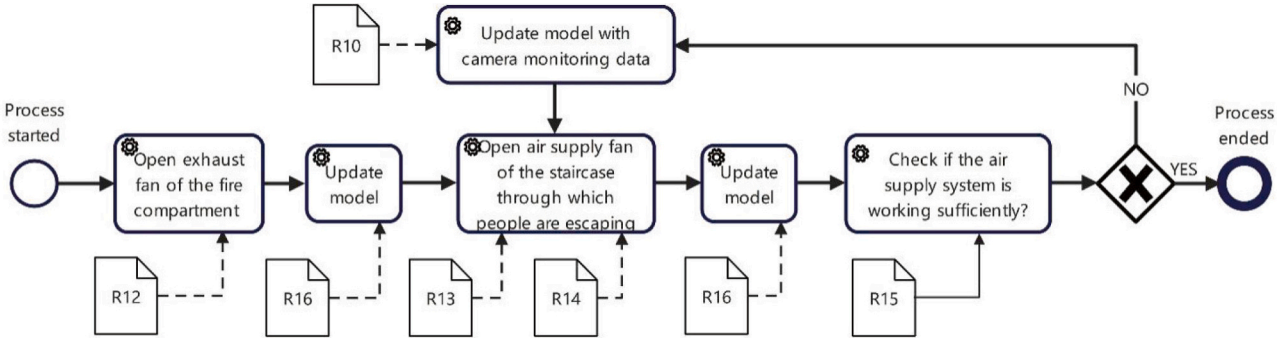


Fig. 19. Process model for intelligent control of smoke control systems.

considered as the prototype of the geometric model. A domain ontology is designed as the semantic model that bridges the information integration between geometric information from BIM and the dynamic information from IoT sensing systems. The integrated information is organised as a DT data model that is considered as a dynamic and precise mapping between the virtual space and the physical space. The process model and the rule model are employed to achieve different intelligent control mechanisms. Within the rule model, the diagnosis rules are whether the intelligent control mechanisms are activated; the process rules specify the response of a fire protection system to a fire emergency; and the update rules are used to keep the virtual space in sync with the physical space. A case study is also investigated in this study to verify the feasibility of the use of the two technologies in handling fire incidents.

Compared with the existing DT framework, this study makes two main contributions. First, the BFP ontology is designed as the semantic model to address the data interoperability between geometric data (e.g., the building model data) and real-time data (e.g., the monitoring data collected by IoT sensors). The BFP ontology is developed according to the idea of ontology reuse and modularity. The concepts and relations in the BFP ontology are developed in accordance with the baseline of the application to more effectively describe the building's geometric, positional, and operating information, as well as the BFP system. Based on the BFP ontology, heterogeneous data and various sources of information can be integrated into a DT data model to support retrieval, reuse, and decision-making. In addition, three types of verification rules, namely, consistency check rules, classification check rules, and composition check rules, are developed to ensure an accurate mapping from the BIM model to the DT data model.

Second, the process model based on BPMN and the rule model are proposed to represent the control mechanism, such that different subsystems can be collaborated in time. Three typical scenarios involving the collaboration between the fire alarm system and the sprinkler

Table 10
Basic rules of rule model for intelligent control of smoke control systems.

No.	Rule type	Representation form	Rule
R11	Diagnosis rule	SPARQL	*ASK {* *?fc rdf:type bfp:FireCompartment .* *{SELECT?fc WHERE {* *?fc bot:hasSpace?space .* *?alarm rdf:type bfp:SmokeDetector .* *?alarm brick:hasLocation?space .* *?alarm bfp:hasRecord?record .* *?record bfp:hasStatus bfp:ALARM .* *?record sosa:resultTime?time .}* *GROUP BY?time?fc HAVING* *(COUNT(?alarm) > 1)* *ORDER BY DESC(?time)* *}* *}*
R12	Process rule	SPARQL	*SELECT?outlet?fan WHERE {* *?fc bot:hasSpace?space .* *?outlet rdf:type bfp:SmokeExhaustOutlet .* *?space brick:feeds?outlet .* *?outlet brick:feeds?fan .* *{SELECT?fc WHERE {* *?fc rdf:type bfp:FireCompartment .* *?fc bot:hasSpace?Fspace .* *?alarm rdf:type bfp:SmokeDetector .* *?alarm brick:hasLocation?Fspace .* *?alarm bfp:hasRecord?Frecord .* *?Frecord bfp:hasStatus bfp:ALARM .* *} GROUP BY?fc HAVING (COUNT(?* *alarm) > 1)* *}* *}*
R13	Process rule	SPARQL	*ASK {* *{SELECT?staircase WHERE {* *VALUES?staircase {bfpi:17301}* *?camera rdf:type bfp:Camera .* *?camera brick:hasLocation?staircase .* *?camera bfp:hasObservation?Observation* *.* *?Observation sosa:hasSimpleResult?* *people .* *?Observation sosa:resultTime?time .* *} GROUP BY?time?staircase* *HAVING (SUM(?people) > 0)* *ORDER BY DESC(?time) LIMIT 1* *}* *}*
R14	Process rule	SPARQL	*SELECT?inlet?fan WHERE {* *?inlet rdf:type bfp:AirSupplyInlet .* *?inlet brick:feeds?staircase .* *?fan brick:feeds?inlet .* *VALUES?staircase {bfpi:17301}* *}*
R15	Process rule	SPARQL	*ASK {?device bfp:hasRecord?record .* *VALUES?device {bfpi:AirSupplyFan_2}* *{SELECT?time?status WHERE {* *?record sosa:resultTime?time .}* *ORDER BY DESC(?time) LIMIT 1}* *?record bfp:hasStatus bfp:ON .* *}*

R16	Update rule	Pseudo code	**Input**: selected devices *ds*; RDF data model *m*; updated time *t*
			Output: Updated RDF data model *m*
			for*d***in***ds***do**
			// add *r* as an individual of *bfp:Record* in *m*
			m.addIndividual(*r*, *bfp:Record*);
			m.add(*r*, *bfp:hasStatus*, *bfp:ON*);
			m.add(*r*, *sosa:resultTime*, *t*);
			m.add(*sp*, *bfp:hasRecord*, *r*);
			end for

system, fire door monitoring system and smoke control system have been investigated to verify the feasibility of the use of the two technologies through FDS simulation. The simulation results indicate that the proposed intelligent control approach can lead to better performance in handling fire accidents. Moreover, the SPARQL query language can help achieve efficient relational retrieval on the DT data model and thus be considered as the basic expression of the diagnosis rules and the process rules in the rule model, as summarised in Table 11. In addition, different integrations of SPARQL vocabularies and various SPARQL query clauses can accomplish different tasks. The update rules are described as pseudo code generally due to the complex update requirements.

The limitations of this study are summarised as follows. First, the proposed approach is only verified by simulation; thus, on-site experiments need to be implemented. Second, this study focuses on the data integration and information-driven intelligent control of a BFP system, including an assumption that the data of the IoT terminals can be fully obtained and well-processed. For instance, the idea of the acquisition of the number of people in a room was inspired by [73], which used deep learning and thermal cameras for human detection in heavy thermal scenarios. This idea also requires further on-site validation and improvement. Third, only three basic fire protection scenarios are investigated. The application of the proposed DT-based framework to achieve the complicated control mechanism of the BFP system should be studied in depth.

Future work can focus on developing a platform unifying the hardware and software to support the implementation and validation of the proposed DT framework for the intelligent control of BFP systems. In addition, the application scope of this study can be promoted from two aspects. First, because the development of the sematic model, process model, and rule model relies on manual work, domain experts can improve the above models to achieve complicated control mechanisms for more fire protection scenarios. Furthermore, it is advised to apply and extend the proposed DT framework to other system control tasks, such as the active control of seismic responses.

6. Conclusions

This paper describes an intelligent control of BFP systems based on DT and semantic web technologies. In general, the implementation of DT faces the issues of cross-domain collaboration and multi-mode and multi-source data interoperability. This study has shown that semantic web technologies, especially the ontology engineering and linked data approaches, can be utilized for the fusion of domain knowledge, BIM information, and IoT sensing data. The integrated data is formed as a DT data model and semantic web technologies can also support efficient information acquisition, extraction, verification, and update. The above findings may be considered as a solid foundation of the implementation of DT.

The second major finding is that a digital thread is necessary for information-driven prediction, logical deduction, and intelligent decision making in DT implementation. The digital thread can be achieved based on the DT data model, a rule model and a process model. The process model, which is executed on a workflow engine, reflects intelligent control mechanisms of BFP systems. Human-machine collaboration is also taken into consideration in the process model to improve the robustness of DT implementation. The rule model is conducted to achieve the activation of the process model and the interaction between the virtual space and the physical space.

Moreover, simulation experiments demonstrate that the proposed intelligent control approach can make rapid response to fire incidents and facilitate human evacuation. To the best of the author's knowledge, this work is the first attempt to explore the potential and feasibility of integrating DT and semantic web technologies in the intelligent control of BFP systems.

Looking forward, further attempts are needed. First, the simulated

SmokeExhaustFan_1 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_2 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_5 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_7 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_1 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_3 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_8 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_4 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
Outlet_6 TURN ON SIGNAL 2022-03-08T13:39:12.633Z
AirSupplyFan_3 TURN ON SIGNAL 2022-03-08T13:40:10.907Z
Inlet_3 TURN ON SIGNAL 2022-03-08T13:40:10.907Z
AirSupplyFan_4 TURN ON SIGNAL 2022-03-08T13:40:10.907Z
Inlet_4 TURN ON SIGNAL 2022-03-08T13:40:10.907Z
AirSupplyFan_1 TURN ON SIGNAL 2022-03-08T13:41:10.936Z
Inlet_1 TURN ON SIGNAL 2022-03-08T13:41:10.936Z
AirSupplyFan_2 TURN ON SIGNAL 2022-03-08T13:41:10.936Z
Inlet_2 TURN ON SIGNAL 2022-03-08T13:41:10.936Z

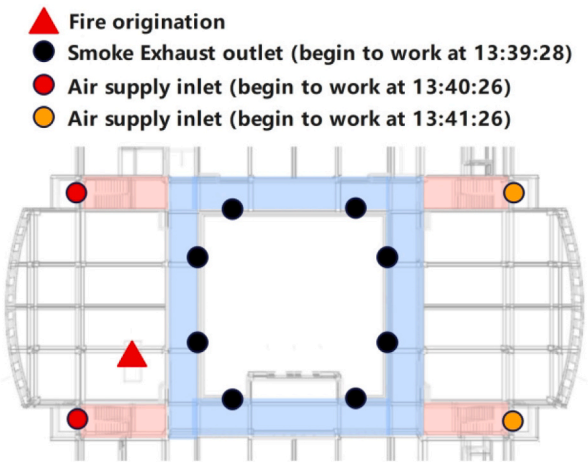


Fig. 20. Start-up time of smoke control systems under intelligent control mechanism.

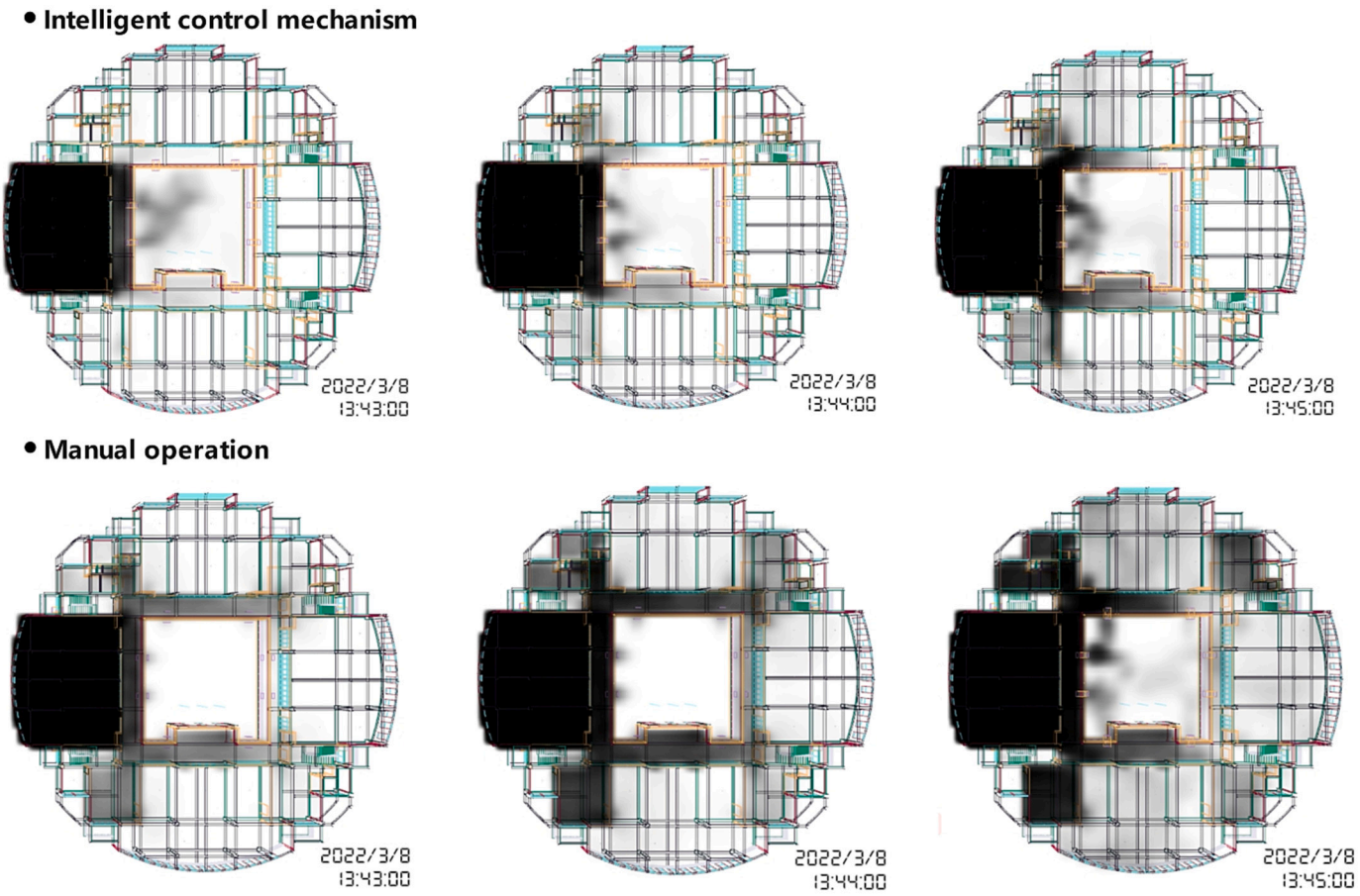


Fig. 21. Smoke distribution under different control mechanisms of smoke control systems.

Table 11
Summary of different rules in rule model.

Rule type	Expression
Diagnosis rule	· Using *CONSTRUCT* clause to obtain the sub data model graph; · Using *ASK* clause to get a Boolean result which is used to decide the activation of the process model.
Process rule	· Using *SELECT* clause to obtain target operating objects; · Using ASK clause to get a Boolean result which is used to decide the continuation of the iteration within the process model; · Using the integration of *GROUP BY*, *HAVING*, *ORDER BY*, and *LIMIT* to get the aggregate results and the latest operating status of the target objects; · Using *FILTER* or *VALUE* to limit the query range.
Update rules	· Normally in described in Pseudo code

experiments are limited to relatively simple scenarios under idealised assumptions. Future research on handling complex intelligent control mechanisms of BFP systems might extend the feasibility and practicability of the proposed approach. Second, given the practical application in engineering projects, it will be important to develop a unified platform and carry out on site experiments. Third, the establishment and improvement of DT models require the involvement of more professional researchers. The development of user-interfacing design tools is in need. Last but not least, this study offers new perspectives for implementing active control of BFP systems in fire accidents. This provides a good starting point for studies in the future. The use of DT and semantic web technologies in achieving active control of building automated systems against emergencies and disasters (e.g., earthquake, terrorist attacks) is desirable for future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgement

This project is funding by the Science and Technology Project ofState Grid Corporation of China, “Research and application of domestic BIM key technologies in power grid engineering” (Grant no. 5200-202156486A-0-5-ZN) .

References

[1] Y. Wang, J. Zheng, B. Ding, Fire protection monitoring system for power cabin in utility tunnel based on IoT technology, in: 2021 IEEE 5th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC), 2021, pp. 1366–1371, https://doi.org/10.1109/ITNEC52019.2021.9586999.
[2] A.M. Kassim, M.M. Roslan, S. Sahak, T.W. Chian, M.A.S.A. Aziz, M.S.H. Basari, M. R. Yaacob, M.A.A. Abid, A.H. Azahar, M.M. Hashim, A.K.R.A. Jaya, T. Yasuno, A. M. Mouazen, Performance evaluation of intelligent fire alarm system with multi data fusion sensor by using IoT platform, in: Proceedings of the Symposium on Intelligent Manufacturing and Mechatronics (SympoSIMM 2021), 2021, pp. 199–209, https://doi.org/10.1007/978-981-16-8954-3_20.
[3] A.O. Arewa, A. Ahmed, D.J. Edwards, C. Nwankwo, Fire safety in high-rise buildings: is the stay-put tactic a misjudgement or magnificent strategy? Buildings 11 (8) (2021) 1–16, https://doi.org/10.3390/buildings11080339.
[4] A.A. Stec, K. Dickens, J.L.J. Barnes, C. Bedford, Environmental contamination following the Grenfell tower fire, Chemosphere 226 (2019) 576–586, https://doi.org/10.1016/j.chemosphere.2019.03.153.
[5] H. Fang, S. Lo, J.T.Y. Lo, Building fire evacuation: an IoT-aided perspective in the 5G era, Buildings 11 (12) (2021) 643, https://doi.org/10.3390/buildings11120643.
[6] J.S. Chou, M.Y. Cheng, Y.M. Hsieh, I.T. Yang, H.T. Hsu, Optimal path planning in real time for dynamic building fire rescue operations using wireless sensors and visual guidance, Autom. Constr. 99 (2019) 1–17, https://doi.org/10.1016/j.autcon.2018.11.020.
[7] Y. Li, J. Yi, X. Zhu, Z. Wang, F. Xu, Developing a fire monitoring and control system based on IoT, in: Proceedings of the 2016 2nd International Conference on Artificial Intelligence and Industrial Engineering (AIIE 2016), 2016, pp. 174–178, https://doi.org/10.2991/aiie-16.2016.40.
[8] H. Chen, L. Hou, G.K. Zhang, S. Moon, Development of BIM, IoT and AR/VR technologies for fire safety and upskilling, Autom. Constr. 125 (2021), 103631, https://doi.org/10.1016/j.autcon.2021.103631.
[9] J.C.P. Cheng, K. Chen, P.K.Y. Wong, W. Chen, C.T. Li, Graph-based network generation and CCTV processing techniques for fire evacuation, Building Research & Information 49 (2021) 179–196, https://doi.org/10.1080/09613218.2020.1759397.
[10] G.B. Ozturk, Digital twin research in the AECO-FM industry, J. Build. Eng. 40 (2021), 102730, https://doi.org/10.1016/j.jobe.2021.102730.
[11] NIC (National Infrastructure Commission), Data for the Public Good, Accessed online, https://nic.org.uk/app/uploads/Data-for-the-Public-Good-NIC-Report.pdf, 2022 (accessed on 27 October 2022).
[12] M.M. Herterich, A. Eck, F. Uebernickel, Exploring how digitized products enable industrial service innovation-an affordance perspective, in: 24rd European Conference on Information Systems (ECIS2016), 2016, pp. 11–15. http://aisel.aisnet.org/ecis2016_rp/156.
[13] Q. Lu, X. Xie, A.K. Parlikad, J.M. Schooling, Digital twin-enabled anomaly detection for built asset monitoring in operation and maintenance, Autom. Constr. 118 (2020), 103277, https://doi.org/10.1016/j.autcon.2020.103277.
[14] W3C, RDF 1.1 Primer, Accessed online, https://www.w3.org/TR/2014/NOTE-rdf11-primer-20140624/, 2022 (accessed on 27 October 2022).
[15] W3C, SPARQL 1.1 Query Language, Accessed online, https://www.w3.org/TR/sparql11-query/, 2022 (accessed on 27 October 2022).
[16] E. González, J.D. Piñeiro, J. Toledo, R. Arnay, L. Acosta, An approach based on the ifcOWL ontology to support indoor navigation, Egypt. Inform. J. 22 (1) (2020) 1–13, https://doi.org/10.1016/j.eij.2020.02.008.
[17] T.R. Gruber, Toward principles for the design of ontologies used for knowledge sharing? International Journal of Human-Computer Studies 43 (5–6) (1995) 907–928, https://doi.org/10.1006/ijhc.1995.1081.
[18] T. Le, H.D. Jeong, Interlinking life-cycle data spaces to support decision making in highway asset management, Autom. Constr. 64 (2016) 54–64, https://doi.org/10.1016/j.autcon.2015.12.016.
[19] M. Venugopal, C.M. Eastman, J. Teizer, An ontology-based analysis of the industry foundation class schema for building information model exchanges, Adv. Eng. Inform. 29 (4) (2015) 940–957, https://doi.org/10.1016/j.aei.2015.09.006.
[20] A. Zeeshan, A. Chimay, D. Ruikar, P. Carrillo, D. Bouchlaghem, Semantic web based services for intelligent mobile construction collaboration, J. Inform. Technol. Construct. 9 (2004) 367–379. https://www.itcon.org/paper/2004/26.
[21] Z. Aziz, C.J. Anumba, D. Ruikar, P. Carrillo, D. Bouchlaghem, Intelligent wireless web services for construction—a review of the enabling technologies, Autom. Constr. 15 (2) (2006) 113–123, https://doi.org/10.1016/j.autcon.2005.03.002.
[22] X. Xu, H. Cai, Semantic approach to compliance checking of underground utilities, Autom. Constr. 109 (2020), 103006, https://doi.org/10.1016/j.autcon.2019.103006.
[23] M. Compton, P. Barnaghi, L. Bermudez, R. García-Castro, O. Corcho, S. Cox, J. Graybeal, M. Hauswirth, C. Henson, A. Herzog, V. Huang, K. Janowicz, W. D. Kelsey, D. Le Phuoc, L. Lefort, M. Leggieri, H. Neuhaus, A. Nikolov, K. Page, A. Passant, A. Sheth, K. Taylor, The SSN ontology of the W3C semantic sensor network incubator group, J. Web Semantics 17 (2022) 25–32, https://doi.org/10.1016/j.websem.2012.05.003.
[24] B. Balaji, A. Bhattacharya, G. Fierro, J. Gao, J. Gluck, D. Hong, A. Johansen, J. Koh, J. Ploennigs, Y. Agarwal, M. Berges, Brick: Towards a unified metadata schema for buildings, in: Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments, 2016, pp. 41–50, https://doi.org/10.1145/2993422.2993577.
[25] M.H. Rasmussen, M. Lefrançois, G.F. Schneider, P. Pauwels, BOT: the building topology ontology of the W3C linked building data group, Semantic Web 12 (2021) 143–161, https://doi.org/10.3233/SW-200385.
[26] M. Kobes, I. Helsloot, B.D. Vries, J.G. Post, Building safety and human behaviour in fire: a literature review, Fire Saf. J. 45 (1) (2010) 1–11, https://doi.org/10.1016/j.firesaf.2009.08.005.
[27] M. Shokouhi, K. Nasiriani, Z. Cheraghi, A. Ardalan, H. Khankeh, H. Fallahzadeh, D. Khorasani-Zavareh, Preventive measures for fire-related injuries and their risk factors in residential buildings: a systematic review, J. Injury Violence Res. 11 (1) (2019) 1–14, https://doi.org/10.5249/jivr.v11i1.1057.
[28] M. Sabbaghzadeh, M. Sheikhkhoshkar, S. Talebi, M. Rezazadeh, M.R. Moghaddam, M. Khanzadi, A BIM-based solution for the optimisation of fire safety measures in the building design, Sustainability 14 (3) (2022) 1626, https://doi.org/10.3390/su14031626.
[29] J. Wang, G. Wei, X. Dong, A dynamic fire escape path planning method with BIM, J. Ambient. Intell. Humaniz. Comput. 12 (11) (2021) 10253–10265, https://doi.org/10.1007/s12652-020-02794-2.
[30] N. Khan, A.K. Ali, S.V. Tran, D. Lee, C. Park, Visual language-aided construction fire safety planning approach in building information modeling, Appl. Sci. 10 (5) (2020) 1704, https://doi.org/10.3390/app10051704.
[31] X. Chen, C. Liu, I. Wu, A BIM-based visualization and warning system for fire rescue, Adv. Eng. Inform. 37 (2018) 42–53, https://doi.org/10.1016/j.aei.2018.04.015.
[32] S. Wang, W. Wang, K. Wang, S. Shih, Applying building information modeling to support fire safety management, Autom. Constr. 59 (2015) 158–167, https://doi.org/10.1016/j.autcon.2015.02.001.
[33] G. Ma, Z. Wu, BIM-based building fire emergency management: combining building users’ behavior decisions, Autom. Constr. 109 (2020), 102975, https://doi.org/10.1016/j.autcon.2019.102975.
[34] P. Johnson, C. Beyler, P. Croce, C. Dubay, M. McNamee, Very early smoke detection apparatus (VESDA), David Packham, John Petersen, Martin Cole: 2017 DiNenno Prize, Fire Sci. Rev. 6 (1) (2017) 5, https://doi.org/10.1186/s40038-017-0019-4.
[35] F. Yang, Z. Cai, L. Su, Y. Xue, X. Chen, Y. Shen, J. Wang, Research on online monitoring and cause identification system of building electrical fire, Int. J. Metrol. Qual. Eng. 13 (2022) 7, https://doi.org/10.1051/ijmqe/2022009.
[36] L M, J J. J S. M E, Designing an IoT based kitchen monitoring and automation system for gas and fire detection, in: 2022 6th International Conference on Computing Methodologies and Communication (ICCMC), 2022, pp.346–353, doi: https://doi.org/10.1109/ICCMC53470.2022.9754118.
[37] T. Han, T. Ma, Z. Fang, Y. Zhang, C. Han, A BIM-IoT and intelligent compaction integrated framework for advanced road compaction quality monitoring and management, Comput. Electr. Eng. 100 (2022), 107981, https://doi.org/10.1016/j.compeleceng.2022.107981.

[38] Z.B. Ismail, Thermal comfort practices for precast concrete building construction projects: towards BIM and IOT integration, Eng. Constr. Archit. Manag. 29 (3) (2022) 1504–1521, https://doi.org/10.1108/ECAM-09-2020-0767.
[39] K.N. Ali, H.H. Alhajlah, M.A. Kassem, Collaboration and risk in building information modelling (BIM): a systematic literature review, Buildings 12 (5) (2022) 571, https://doi.org/10.3390/buildings12050571.
[40] M. Shafto, M. Conroy, R. Doyle, E. Glaessgen, C. Kemp, J. LeMoigne, L. Wang, Modeling, Simulation, Information Technology & Processing Roadmap, Technology Area 11, Washington, DC: National Aeronautics and Space Administration, Accessed online: https://www.nasa.gov/sites/default/files/501321main_TA11-ID_rev4_NRC-wTASR.pdf, 2022 (accessed on 27 October 2022).
[41] F. Tao, J. Cheng, Q. Qi, M. Zhang, H. Zhang, F. Sui, Digital twin-driven product design, manufacturing and service with big data, Int. J. Adv. Manuf. Technol. 94 (2018) 3563–3576, https://doi.org/10.1007/s00170-017-0233-1.
[42] A. Bolton, M. Enzer, J. Schooling, The Gemini Principles, Accessed online, https://www.cdbb.cam.ac.uk/system/files/documents/TheGeminiPrinciples.pdf, 2022 (accessed on 27 October 2022).
[43] H. Liang, N. Mulatibieke, J. Shi, Z. Lv, L. Zhou, Application of digital twin in operation and maintenance of power transmission project, in: 2021 Power System and Green Energy Conference (PSGEC), 2021, pp. 577–582, https://doi.org/10.1109/PSGEC51302.2021.9541721.
[44] L. Zhou, C. An, J. Shi, Z. Lv, H. Liang, Design and construction integration technology based on digital twin, in: 2021 Power System and Green Energy Conference (PSGEC), 2021, pp. 7–11, https://doi.org/10.1109/PSGEC51302.2021.9541682.
[45] Y. Zhao, C. Cao, Z. Liu, A framework for prefabricated component hoisting management systems based on digital twin technology, Buildings 12 (2022) 276, https://doi.org/10.3390/buildings12030276.
[46] Q. Lu, A.K. Parlikad, P. Woodall, G.D. Ranasinghe, Developing a digital twin at building and city levels: case study of West Cambridge campus, J. Manag. Eng. 36 (2020) 5020004, https://doi.org/10.1061/(ASCE)ME.1943-5479.0000763.
[47] S. Agostinelli, F. Cumo, G. Guidi, C. Tomazzoli, Cyber-physical systems improving building energy management: digital twin and artificial intelligence, Energies 14 (8) (2021) 2338, https://doi.org/10.3390/en14082338.
[48] V. Stojanovic, M. Trapp, R. Richter, B. Hagedorn, J. Döllner, Towards the generation of digital twins for facility management based on 3D point clouds, in: 34th Annual Association of Researchers in Construction Management Conference (ARCOM 2018), 2018, pp. 270–279, in: http://www.arcom.ac.uk/-docs/proceedings/b65e593d342a8de045cf05698677e600.pdf.
[49] R. Lu, I. Brilakis, Digital twinning of existing reinforced concrete bridges from labelled point clusters, Autom. Constr. 105 (2019), 102837, https://doi.org/10.1016/j.autcon.2019.102837.
[50] Z. Liu, G. Shi, X. Meng, Z. Sun, Intelligent control of building operation and maintenance processes based on global navigation satellite system and digital twins, Remote Sens. 14 (6) (2022) 1387, https://doi.org/10.3390/rs14061387.
[51] Z. Liu, A. Li, Z. Sun, G. Shi, X. Meng, Digital twin-based risk control during prefabricated building hoisting operations, Sensors 22 (7) (2022) 2522, https://doi.org/10.3390/s22072522.
[52] K. Kluza, K. Honkisz, From SBVR to BPMN and DMN models, proposal of translation from rules to process and decision models, in: Proceedings of International Conference on Artificial Intelligence and Soft Computing (ICAISC 2016), 2016, pp. 453–462, https://doi.org/10.1007/978-3-319-39384-1_39.
[53] Object Management Group (OMG), Business Process Model and Notation (BPMN) Version 2.0, Accessed online, https://www.omg.org/spec/BPMN/2.0/PDF, 2022 (accessed on 27 October 2022).
[54] D. Mavrokapnidis, K. Katsigarakis, P. Pauwels, E. Petrova, I. Korolija, D. Rovas, A linked-data paradigm for the integration of static and dynamic building data in Digital Twins, in: Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2021, pp. 369–372, https://doi.org/10.1145/3486611.3491125.
[55] F.Z. Amara, M. Hemam, M. Djezzar, M. Maimour, Semantic web technologies for internet of things semantic interoperability, in: the International Conference on Information, Communication & Cybersecurity, 2021, pp. 133–143, https://doi.org/10.1007/978-3-030-91738-8_13.
[56] W3C, SWRL: A Semantic Web Rule Language Combining OWL and RuleML, Accessed online, https://www.w3.org/Submission/SWRL/, 2022 (accessed on 27 October 2022).
[57] Apache Jena, Reasoners and Rule Engines: Jena Inference Support, Accessed online, https://jena.apache.org/documentation/inference/index.html, 2022 (accessed on 27 October 2022).
[58] W3C, RDF Schema 1.1, Accessed online, https://www.w3.org/TR/2014/REC-rdf-schema-20140225/, 2022 (accessed on 27 October 2022).
[59] W3C, OWL 2 Web Ontology Language Primer (Second Edition), Accessed online, https://www.w3.org/TR/2012/REC-owl2-primer-20121211/, 2022 (accessed on 27 October 2022).
[60] L. Jiang, J. Shi, C. Wang, Multi-ontology fusion and rule development to facilitate automated code compliance checking using BIM and rule-based reasoning, Adv. Eng. Inform. 51 (2022), 101449, https://doi.org/10.1016/j.aei.2021.101449.
[61] B. Zhong, C. Gan, H. Luo, X. Xing, Ontology-based framework for building environmental monitoring and compliance checking under BIM environment, Build. Environ. 141 (2018) 127–142, https://doi.org/10.1016/j.buildenv.2018.05.046.
[62] W. Terkaj, G.F. Schneider, P. Pauwels, Reusing domain ontologies in linked building data: The case of building automation and control, in: The 8th International Workshop on Formal Ontologies Meet Industry, 2017. http://hdl.handle.net/1854/LU-8578705.
[63] S. Hu, J. Wang, C. Hoare, Y. Li, P. Pauwels, J. O'Donnell, Building energy performance assessment using linked data and cross-domain semantic reasoning, Autom. Constr. 124 (2021), 103580, https://doi.org/10.1016/j.autcon.2021.103580.
[64] M.H. Rasmussen, M. Lefrançois, P. Pauwels, C.A. Hviid, J. Karlshøj, Managing interrelated project information in AEC knowledge graphs, Autom. Constr. 108 (2019), 102956, https://doi.org/10.1016/j.autcon.2019.102956.
[65] H. Gao, B. Zhong, H. Luo, W. Chen, Computational geometric approach for BIM semantic enrichment to support automated underground garage compliance checking, J. Constr. Eng. Manag. 148 (2022) 05021013, https://doi.org/10.1061/(ASCE)CO.1943-7862.0002230.
[66] P. Pauwels, W. Terkaj, EXPRESS to OWL for construction industry: towards a recommendable and usable ifcOWL ontology, Autom. Constr. 63 (2016) 100–133, https://doi.org/10.1016/j.autcon.2015.12.003.
[67] N.F. Noy, D.L. McGuinness, Ontology Development 101: A Guide to Creating your First Ontology, Accessed online, https://redirect.cs.umbc.edu/courses/graduate/691/fall17/01/papers/ontology101.pdf, 2022.
[68] IfcOpenShell, The Open Source ifc Toolkit and Geometry Engine, Accessed online, http://www.ifcopenshell.org/, 2022 (accessed on 27 October 2022).
[69] L. Tanklevskiy, A. Tsoy, A. Snegirev, Electrically controlled dynamic sprinkler activation: computational assessment of potential efficiency, Fire Saf. J. 91 (2017) 614–623, https://doi.org/10.1016/j.firesaf.2017.04.019.
[70] M. Li, S. Zhu, J. Wang, Z. Zhou, Research on fire safety evacuation in a university library in Nanjing, Procedia Engineering 211 (2018) 372–378, https://doi.org/10.1016/j.proeng.2017.12.025.
[71] W. Wang, H. Zhang, Influence of smoke exhaust start-up time on smoke control, Journal of Tsinghua University (Science & Technology) 56 (2016) 223–228, https://doi.org/10.16511/j.cnki.qhdxxb.2015.22.023.
[72] A. Bochkovskiy, C.Y. Wang, H.M. Liao, YOLOv4: Optimal Speed and Accuracy of Object Detection, Accessed online, https://arxiv.org/abs/2004.10934, 2022.
[73] P. Tsai, C. Liao, S. Yuan, Using deep learning with thermal imaging for human detection in heavy smoke scenarios, Sensors 22 (14) (2022) 5351, https://doi.org/10.3390/s22145351.